

Project Management

3



Outline

- ▶ Global Company Profile:
Bechtel Group
- ▶ The Importance of Project Management
- ▶ Project Planning
- ▶ Project Scheduling
- ▶ Project Controlling

Outline - Continued

- ▶ Project Management Techniques: PERT and CPM
- ▶ Determining the Project Schedule
- ▶ Variability in Activity Times
- ▶ Cost-Time Trade-offs and Project Crashing

Outline - Continued

- ▶ A Critique of PERT and CPM
- ▶ Using Microsoft Project to Manage Projects

Learning Objectives

When you complete this chapter you should be able to:

- 1. Use** a Gantt chart for scheduling
- 2. Draw** AOA and AON networks
- 3. Complete** forward and backward passes for a project
- 4. Determine** a critical path

Learning Objectives

When you complete this chapter you should be able to:

- 5. Calculate** the variance of activity times
- 6. Crash** a project

Bechtel Projects

- ▶ Constructing 30 high-security data centers worldwide for Equinix, Inc. (\$1.2 billion)
- ▶ Building and running a rail line between London and the Channel Tunnel (\$4.6 billion)
- ▶ Developing an oil pipeline from the Caspian Sea region to Russia (\$850 million)
- ▶ Expanding the Dubai Airport in the UAE (\$600 million), and the Miami Airport in Florida (\$2 billion)

Bechtel Projects

- ▶ Building liquid natural gas plants in Yemen (\$2 billion) and in Trinidad, West Indies (\$1 billion)
- ▶ Building a new subway for Athens, Greece (\$2.6 billion)
- ▶ Constructing a natural gas pipeline in Thailand (\$700 million)
- ▶ Building 30 plants for iMotors.com, a company that sells refurbished autos online (\$300 million)
- ▶ Building a highway to link the north and south of Croatia (\$303 million)

Importance of Project Management

- ▶ Bechtel Project Management
 - ▶ International workforce, construction professionals, cooks, medical personnel, security
 - ▶ Strategic value of time-based competition
 - ▶ Quality mandate for continual improvement

Project Characteristics

- ▶ Single unit
- ▶ Many related activities
- ▶ Difficult production planning and inventory control
- ▶ General purpose equipment
- ▶ High labor skills

Examples of Projects

- ▶ Building Construction



- ▶ Research Project

Management of Projects

1. *Planning* - goal setting, defining the project, team organization
2. *Scheduling* - relate people, money, and supplies to specific activities and activities to each other
3. *Controlling* - monitor resources, costs, quality, and budgets; revise plans and shift resources to meet time and cost demands

Project Management Activities

- ▶ Planning

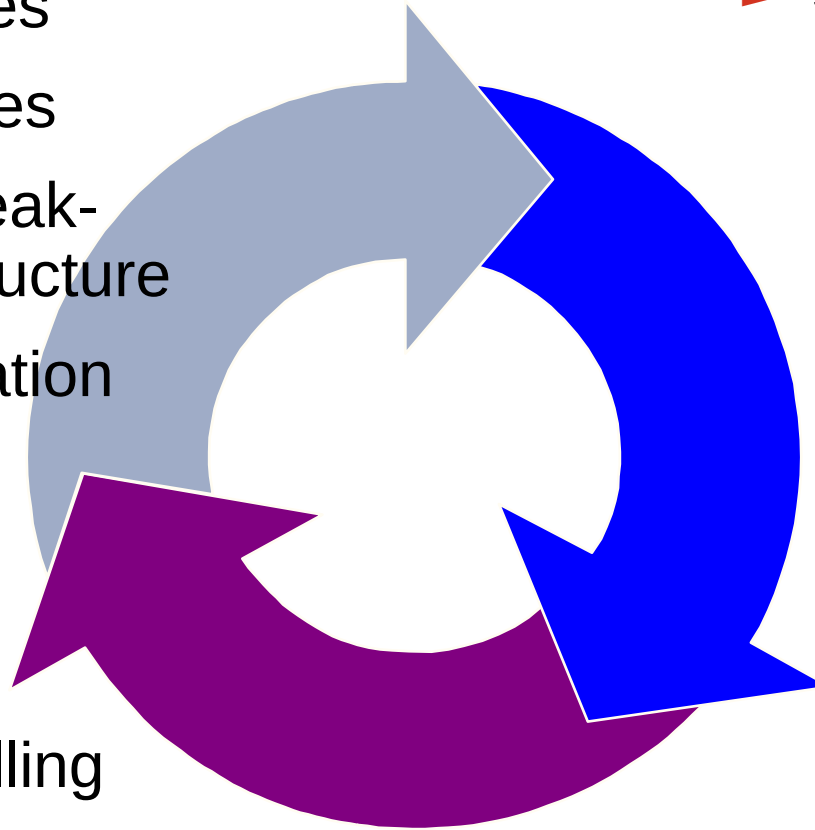
- ▶ Objectives
- ▶ Resources
- ▶ Work break-down structure
- ▶ Organization

- ▶ Scheduling

- ▶ Project activities
- ▶ Start & end times
- ▶ Network

- ▶ Controlling

- ▶ Monitor, compare, revise, action



Project Planning, Scheduling, and Controlling

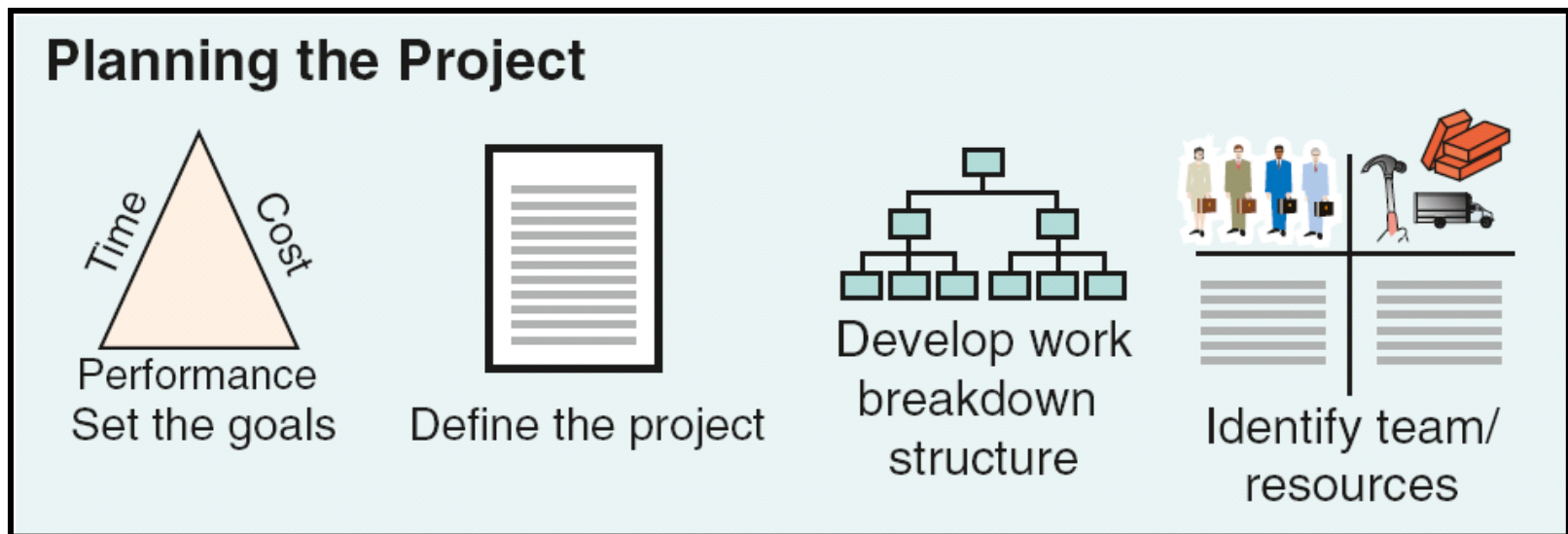


Figure 3.1

Project Planning, Scheduling, and Controlling

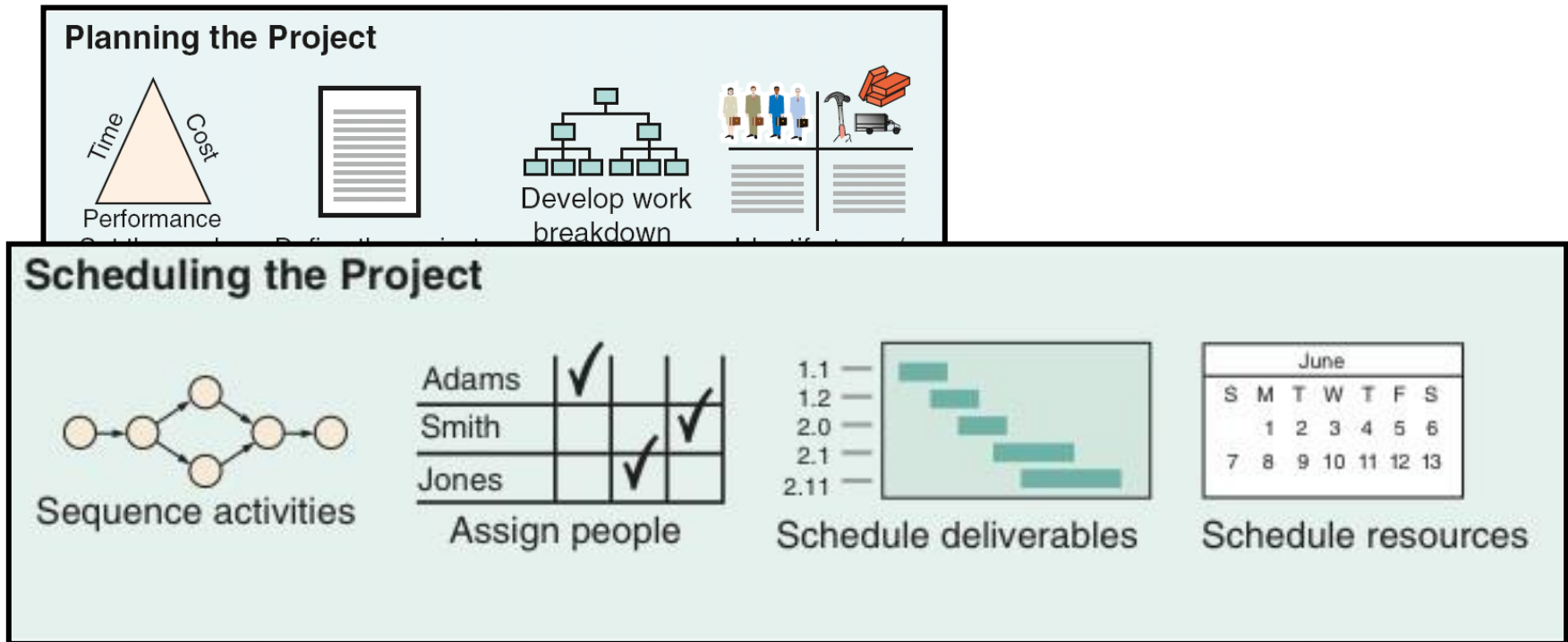


Figure 3.1

Project Planning, Scheduling, and Controlling

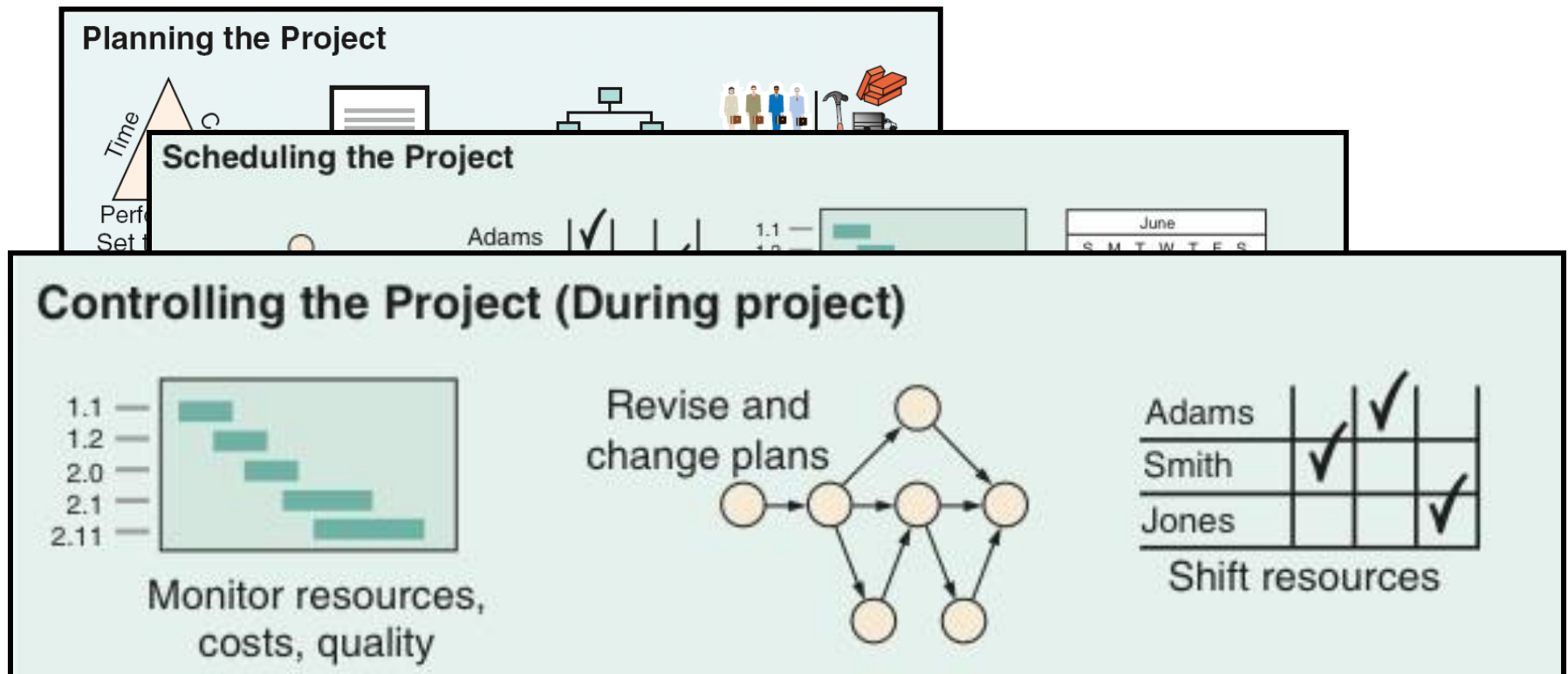


Figure 3.1

Project Planning, Scheduling, and Controlling

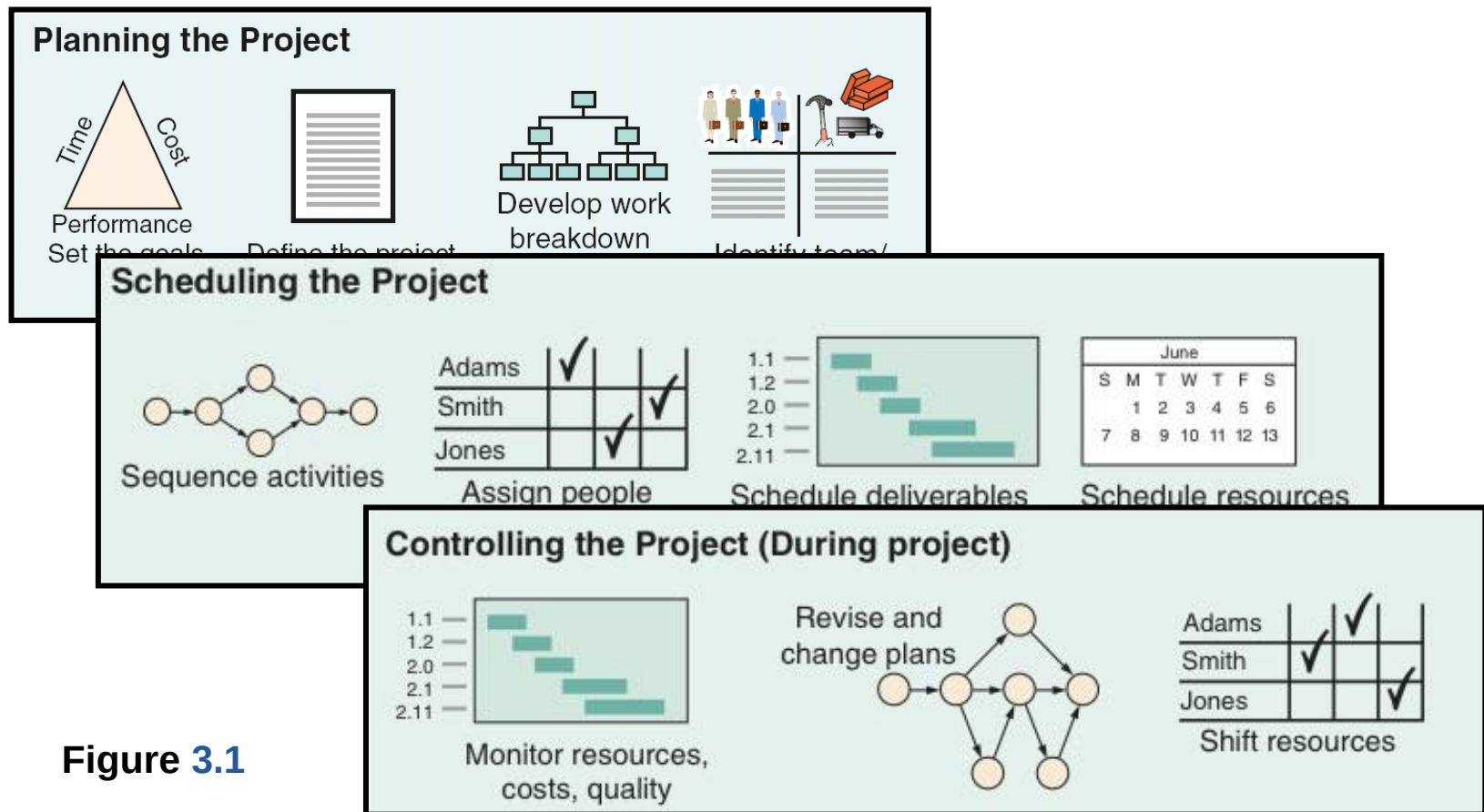
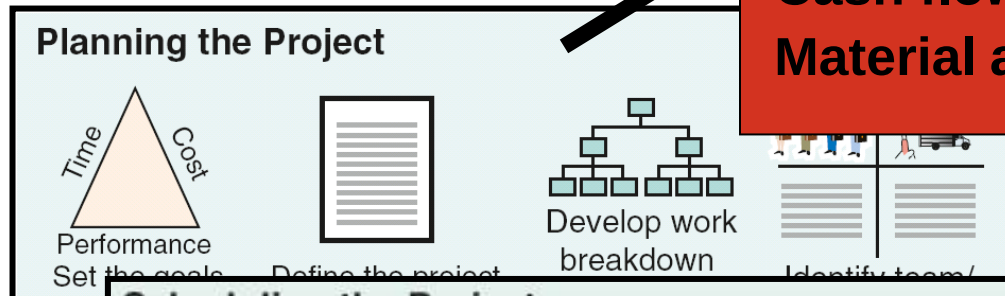


Figure 3.1

Project Scheduling, Planning

Time/cost estimates
Budgets
Engineering diagrams
Cash flow charts
Material availability details



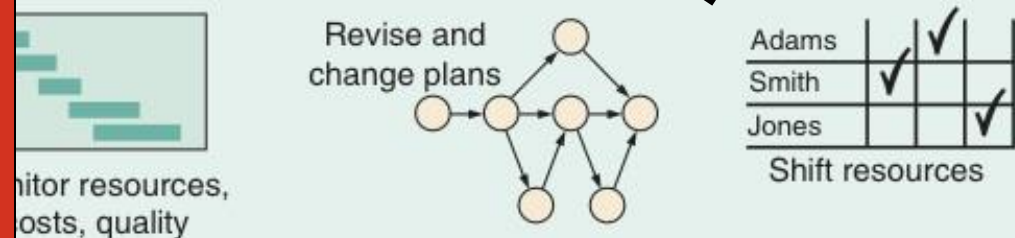
Scheduling the Project



Budgets
Delayed activities report
Slack activities report

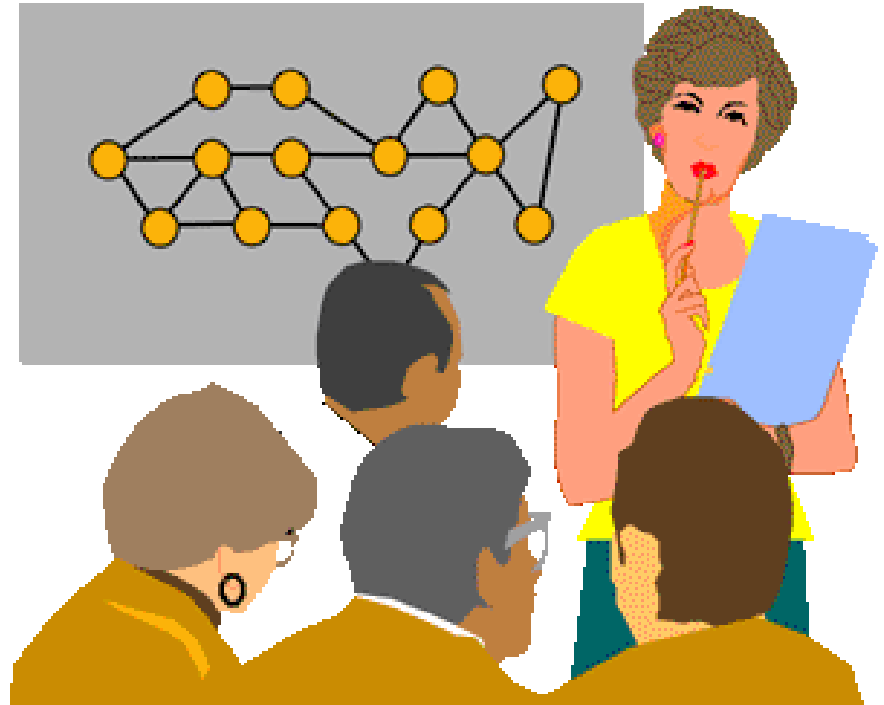
CPM/PERT
Gantt charts
Milestone charts
Cash flow schedules

Monitoring the Project (During project)



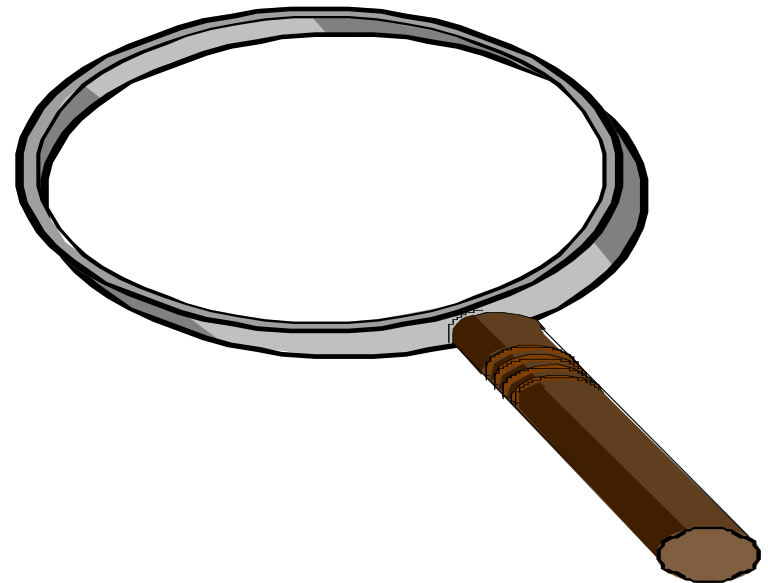
Project Planning

- ▶ Establishing objectives
- ▶ Defining project
- ▶ Creating work breakdown structure
- ▶ Determining resources
- ▶ Forming organization



Project Organization

- ▶ Often temporary structure
- ▶ Uses specialists from entire company
- ▶ Headed by project manager
 - ▶ Coordinates activities
 - ▶ Monitors schedule and costs
- ▶ Permanent structure called 'matrix organization'



Project Organization Works Best When

1. Work can be defined with a specific goal and deadline
2. The job is unique or somewhat unfamiliar to the existing organization
3. The work contains complex interrelated tasks requiring specialized skills
4. The project is temporary but critical to the organization
5. The project cuts across organizational lines

A Sample Project Organization

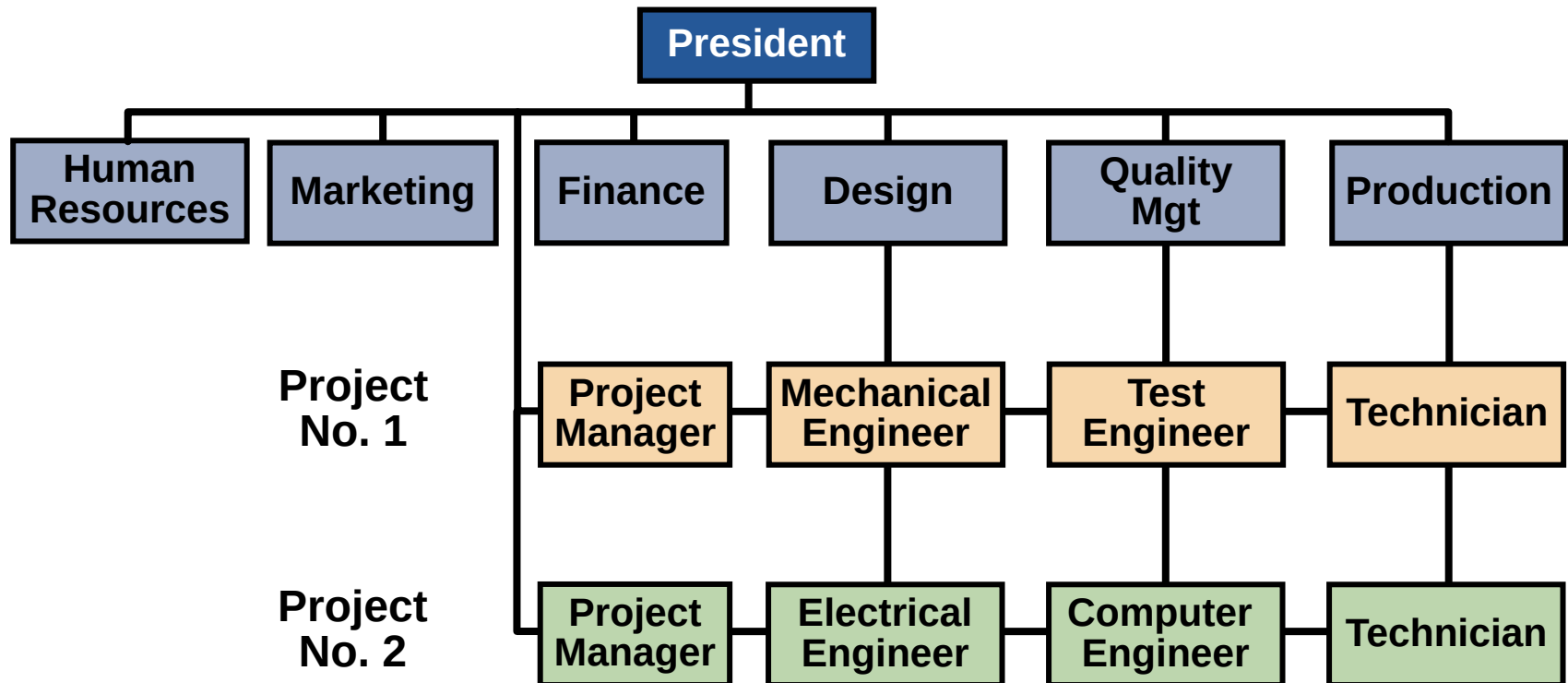
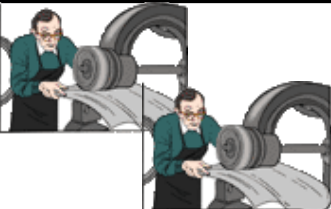


Figure 3.2

Matrix Organization

	Marketing	Operations	Engineering	Finance
Project 1				
Project 2				
Project 3				
Project 4				

The Role of the Project Manager

Highly visible

Responsible for making sure that:

1. All necessary activities are finished in order and on time
2. The project comes in within budget
3. The project meets quality goals
4. The people assigned to the project receive motivation, direction, and information

The Role of the Project Manager

Highly visible
Responsible for

1. All necessary activities completed and on time
2. The project completed on time
3. The project meeting the requirements
4. The people assigned to the project receive motivation, direction, and information

Project managers should be:

- ▶ **Good coaches**
- ▶ **Good communicators**
- ▶ **Able to organize activities from a variety of disciplines**

Ethical Issues

- ▶ Project managers face many ethical decisions on a daily basis
- ▶ The Project Management Institute has established an ethical code to deal with problems such as:
 1. Offers of gifts from contractors
 2. Pressure to alter status reports to mask delays
 3. False reports for charges of time and expenses
 4. Pressure to compromise quality to meet schedules

Work Breakdown Structure

Level

1. Project
2. Major tasks in the project
3. Subtasks in the major tasks
4. Activities (or “work packages”) to be completed

Work Breakdown Structure

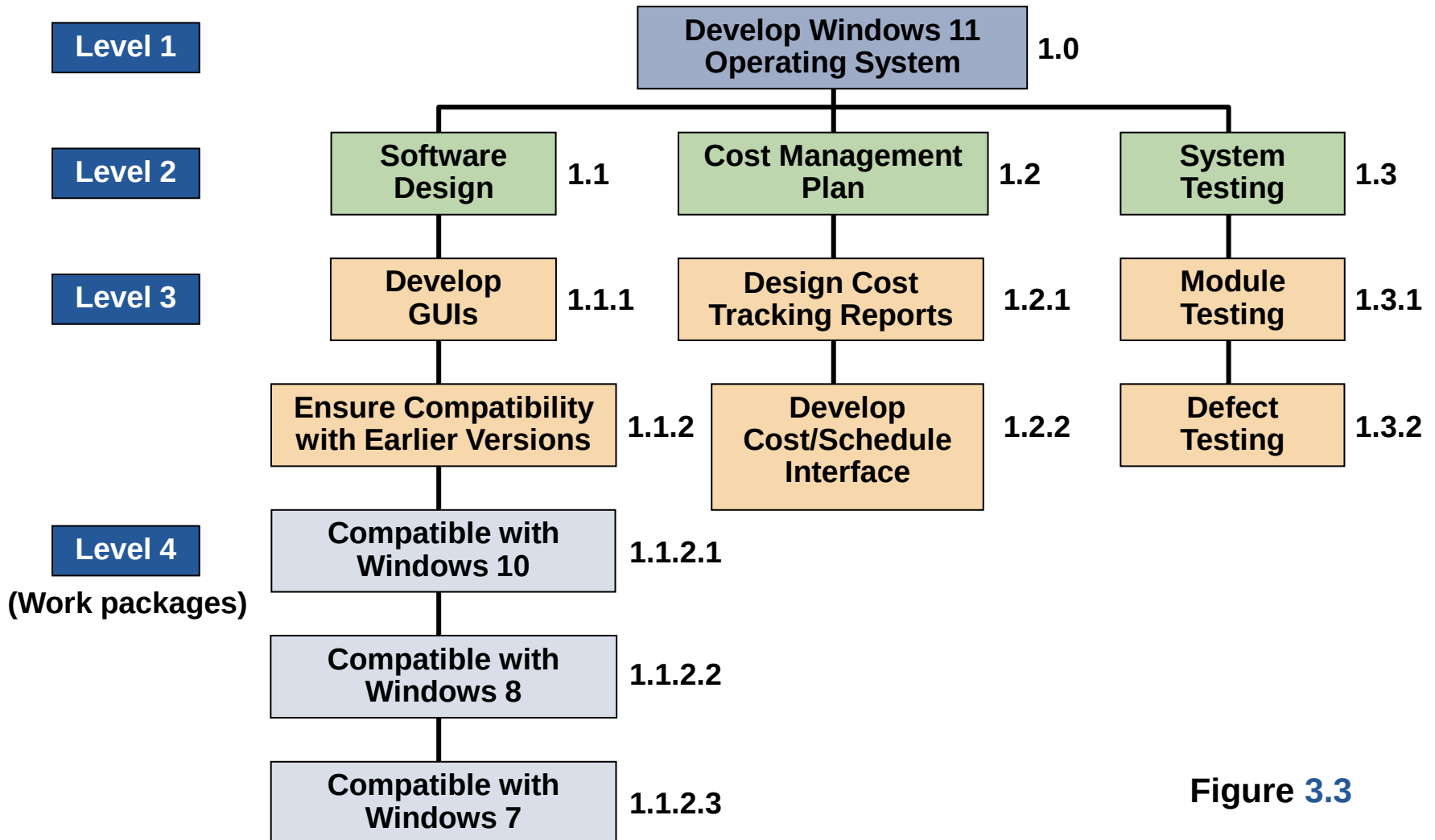


Figure 3.3

Project Scheduling Techniques

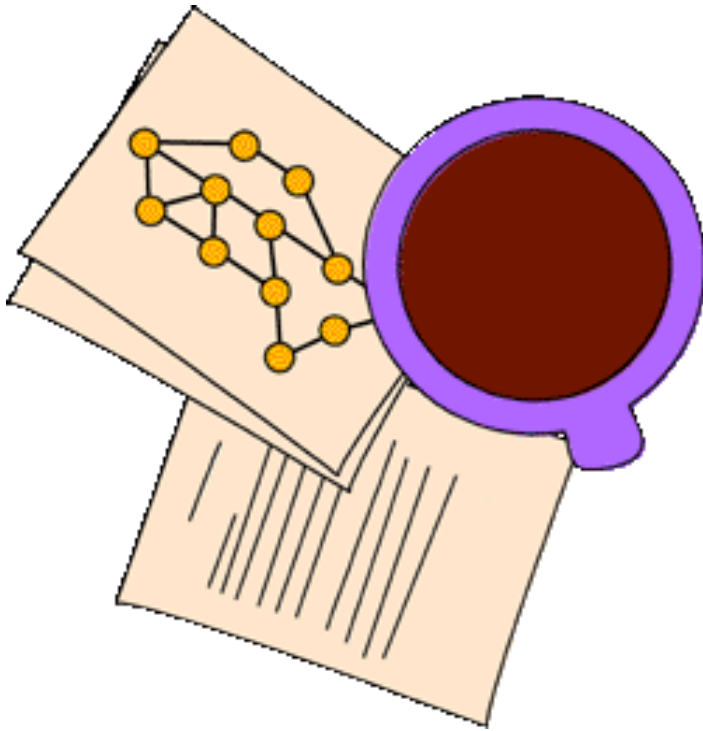
1. Ensure that all activities are planned for
2. Their order of performance is accounted for
3. The activity time estimates are recorded
4. The overall project time is developed



Purposes of Project Scheduling

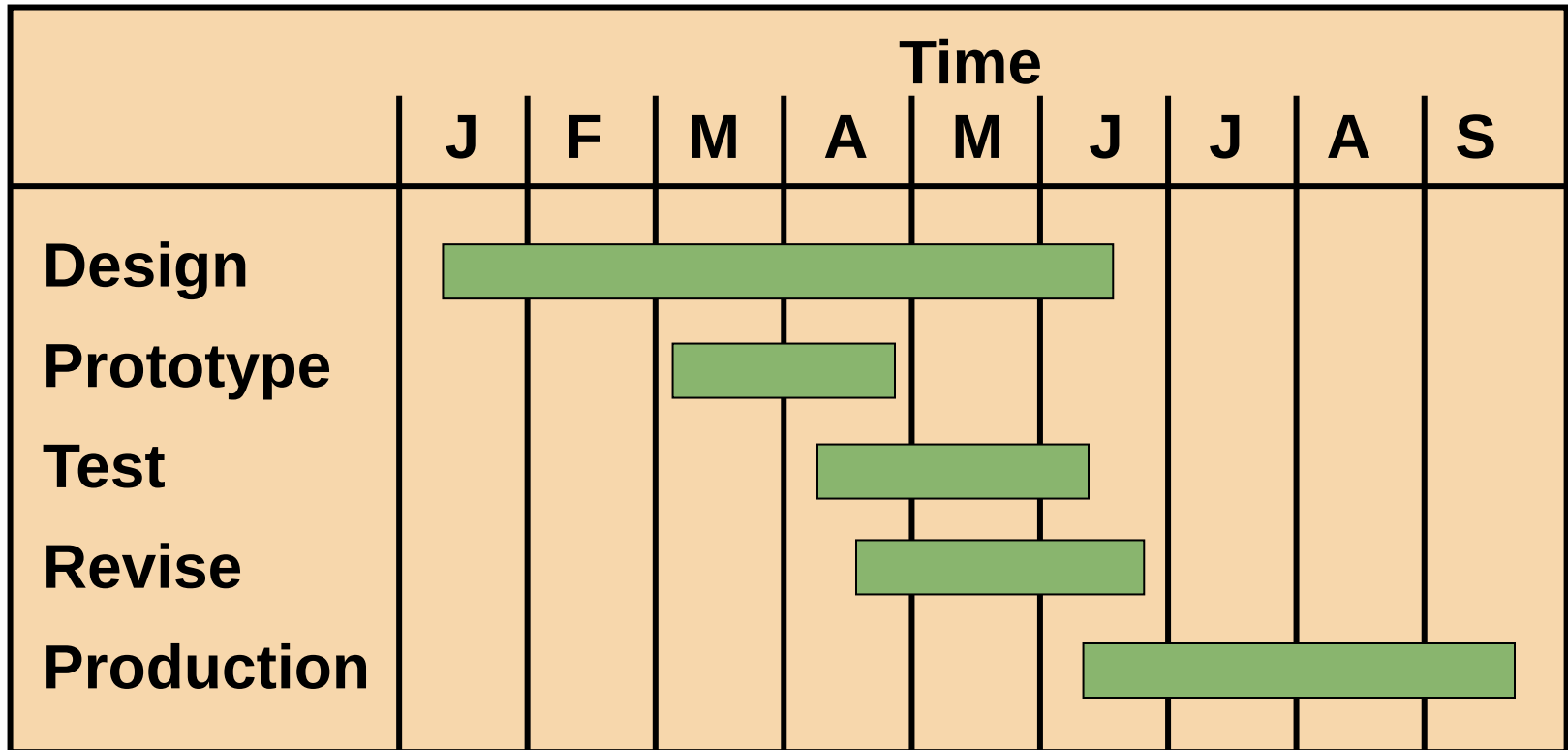
1. Shows the relationship of each activity to others and to the whole project
2. Identifies the precedence relationships among activities
3. Encourages the setting of realistic time and cost estimates for each activity
4. Helps make better use of people, money, and material resources by identifying critical bottlenecks in the project

Project Management Techniques



- ▶ Gantt chart
- ▶ Critical Path Method (CPM)
- ▶ Program Evaluation and Review Technique (PERT)

A Simple Gantt Chart



Service For a Delta Jet

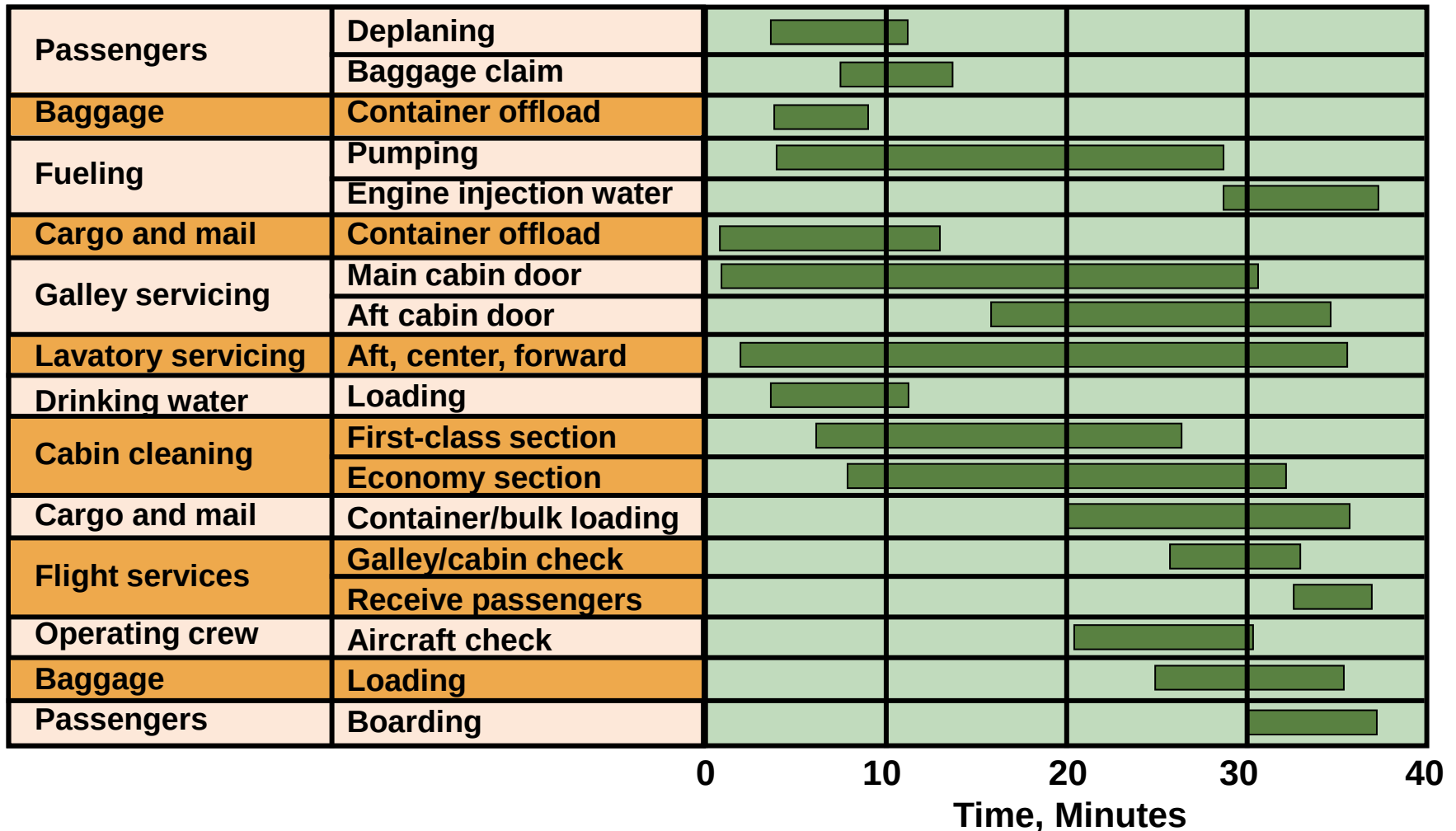


Figure 3.4

Project Controlling

- ▶ Close monitoring of resources, costs, quality, budgets
- ▶ Feedback enables revising the project plan and shift resources
- ▶ Computerized tools produce extensive reports



Project Management Software

- ▶ There are several popular packages for managing projects
 - ▶ Primavera
 - ▶ MacProject
 - ▶ MindView
 - ▶ HP Project
 - ▶ Fast Track
 - ▶ Microsoft Project

Project Control Reports

- ▶ Detailed cost breakdowns for each task
- ▶ Total program labor curves
- ▶ Cost distribution tables
- ▶ Functional cost and hour summaries
- ▶ Raw materials and expenditure forecasts
- ▶ Variance reports
- ▶ Time analysis reports
- ▶ Work status reports

PERT and CPM

- ▶ Network techniques
- ▶ Developed in 1950s
 - ▶ CPM by DuPont for chemical plants (1957)
 - ▶ PERT by Booz, Allen & Hamilton with the U.S. Navy, for Polaris missile (1958)
- ▶ Consider precedence relationships and interdependencies
- ▶ Each uses a different estimate of activity times

Six Steps PERT & CPM

1. Define the project and prepare the work breakdown structure
2. Develop relationships among the activities - decide which activities must precede and which must follow others
3. Draw the network connecting all of the activities

Six Steps PERT & CPM

4. Assign time and/or cost estimates to each activity

5. Compute the longest time path through the network – this is called the critical path

6. Use the network to help plan, schedule, monitor, and control the project

Questions PERT & CPM Can Answer

1. When will the entire project be completed?
2. What are the critical activities or tasks in the project?
3. Which are the noncritical activities?
4. What is the probability the project will be completed by a specific date?

Questions PERT & CPM Can Answer

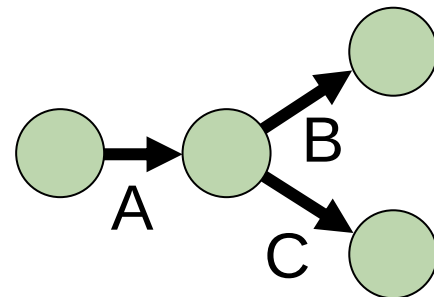
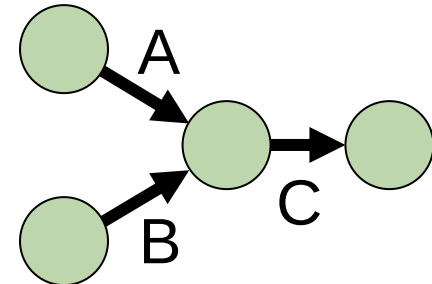
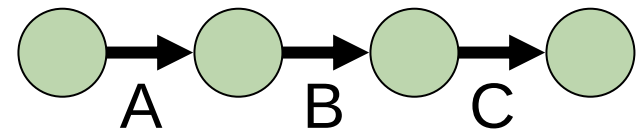
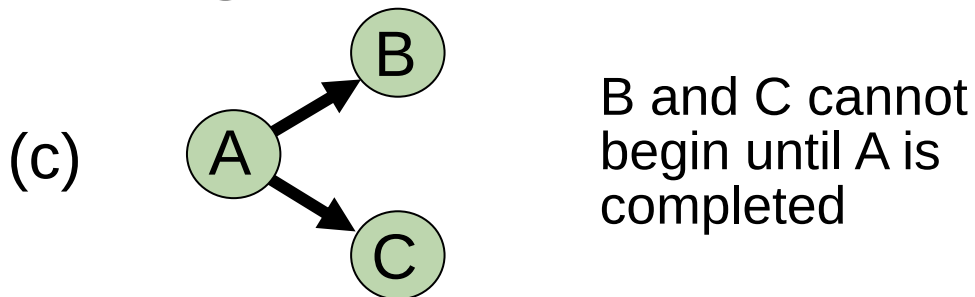
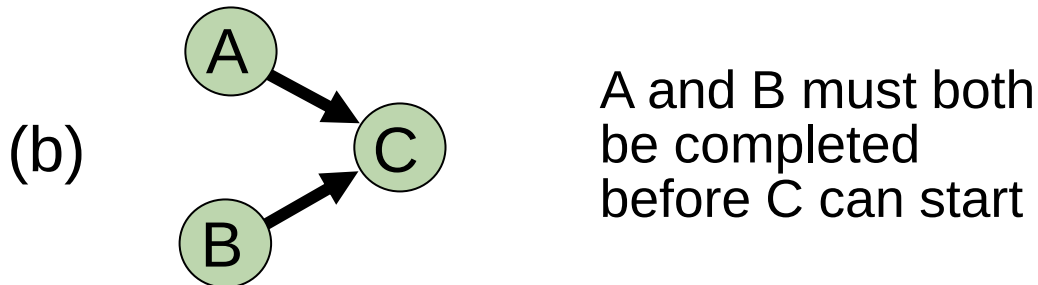
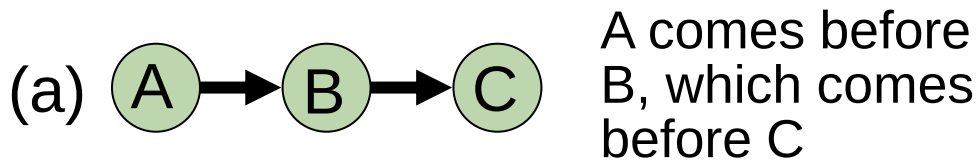
5. Is the project on schedule, behind schedule, or ahead of schedule?
6. Is the money spent equal to, less than, or greater than the budget?
7. Are there enough resources available to finish the project on time?
8. If the project must be finished in a shorter time, what is the way to accomplish this at least cost?

A Comparison of AON and AOA Network Conventions

**Activity on
Node (AON)**

**Activity
Meaning**

**Activity on
Arrow (AOA)**



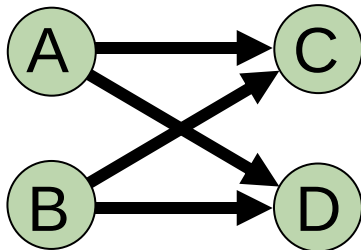
A Comparison of AON and AOA Network Conventions

Activity on
Node (AON)

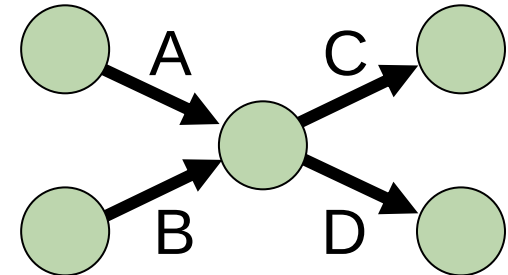
Activity
Meaning

Activity on
Arrow (AOA)

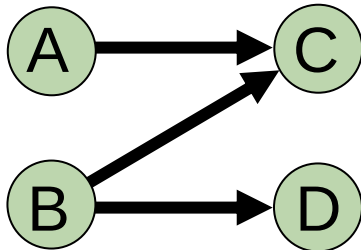
(d)



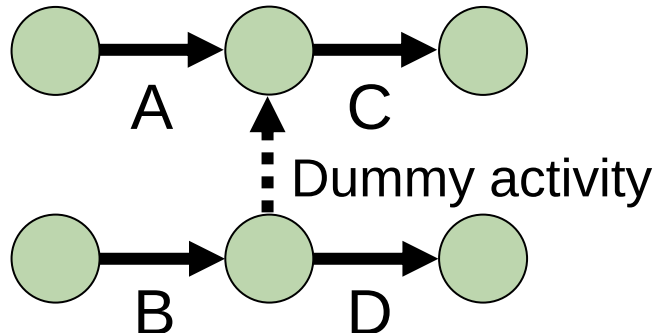
C and D cannot
begin until both
A and B are
completed



(e)



C cannot begin until
both A and B are
completed
D cannot begin until
B is completed
A dummy activity is
introduced in AOA



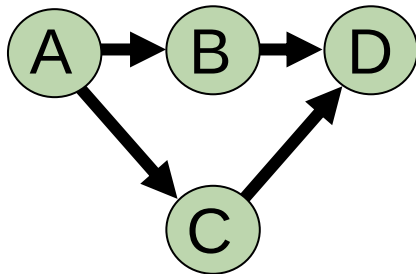
A Comparison of AON and AOA Network Conventions

Activity on
Node (AON)

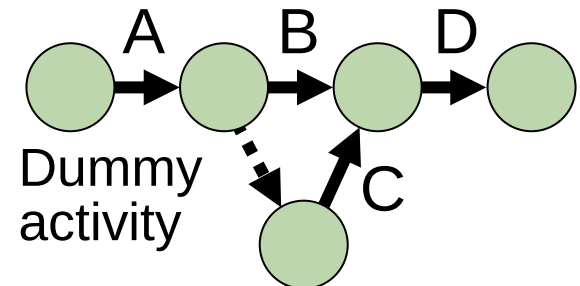
Activity
Meaning

Activity on
Arrow (AOA)

(f)



B and C cannot
begin until A is
completed
D cannot begin
until both B and C
are completed
A dummy activity
is again
introduced in AOA



AON Example

Table 3.1 Milwaukee Paper Manufacturing's Activities and Predecessors

ACTIVITY	DESCRIPTION	IMMEDIATE PREDECESSORS
A	Build internal components	—
B	Modify roof and floor	—
C	Construct collection stack	A
D	Pour concrete and install frame	A, B
E	Build high-temperature burner	C
F	Install pollution control system	C
G	Install air pollution device	D, E
H	Inspect and test	F, G

AON Network for Milwaukee Paper

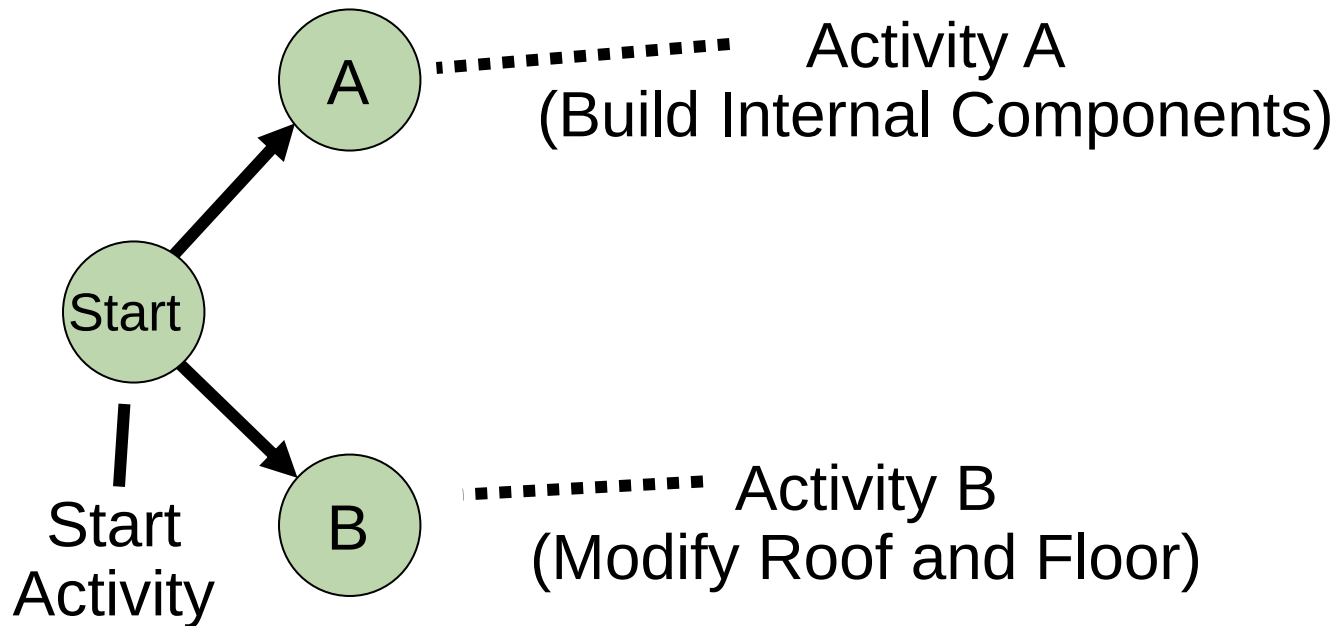


Figure 3.5

AON Network for Milwaukee Paper

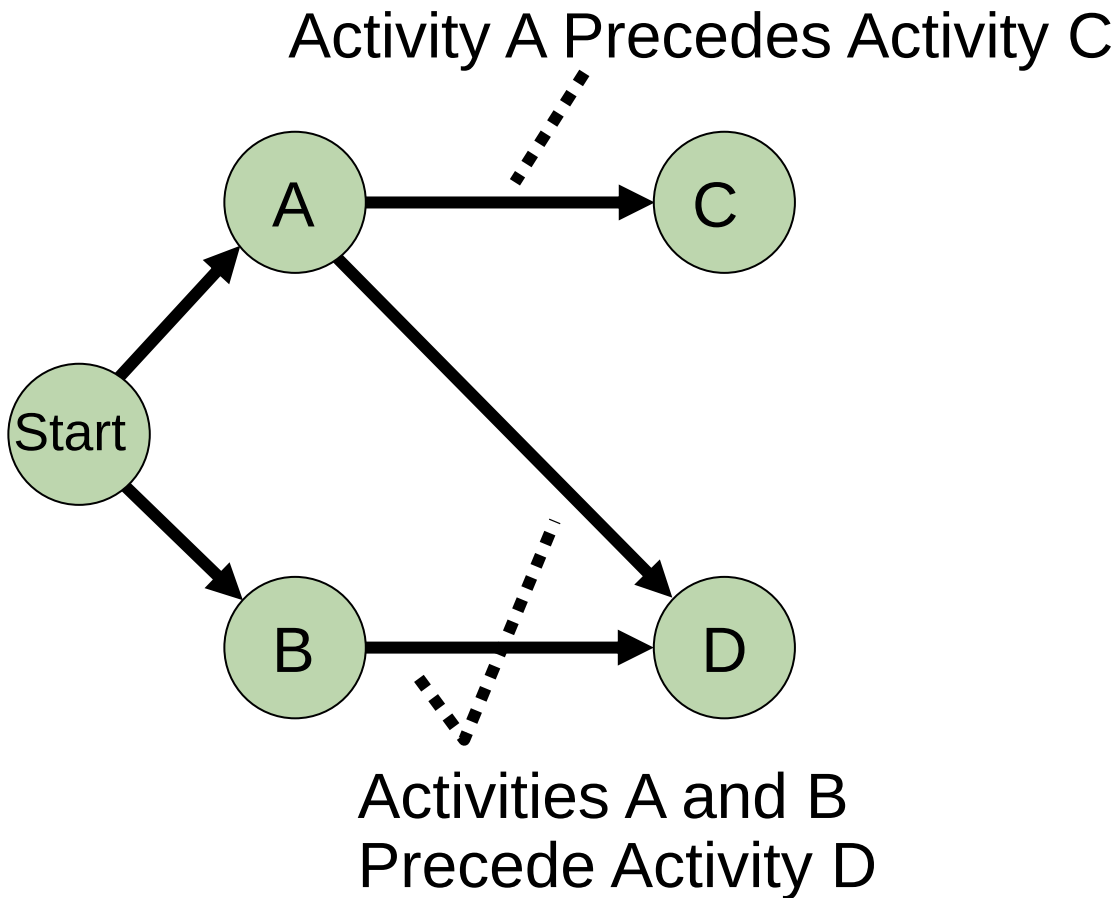
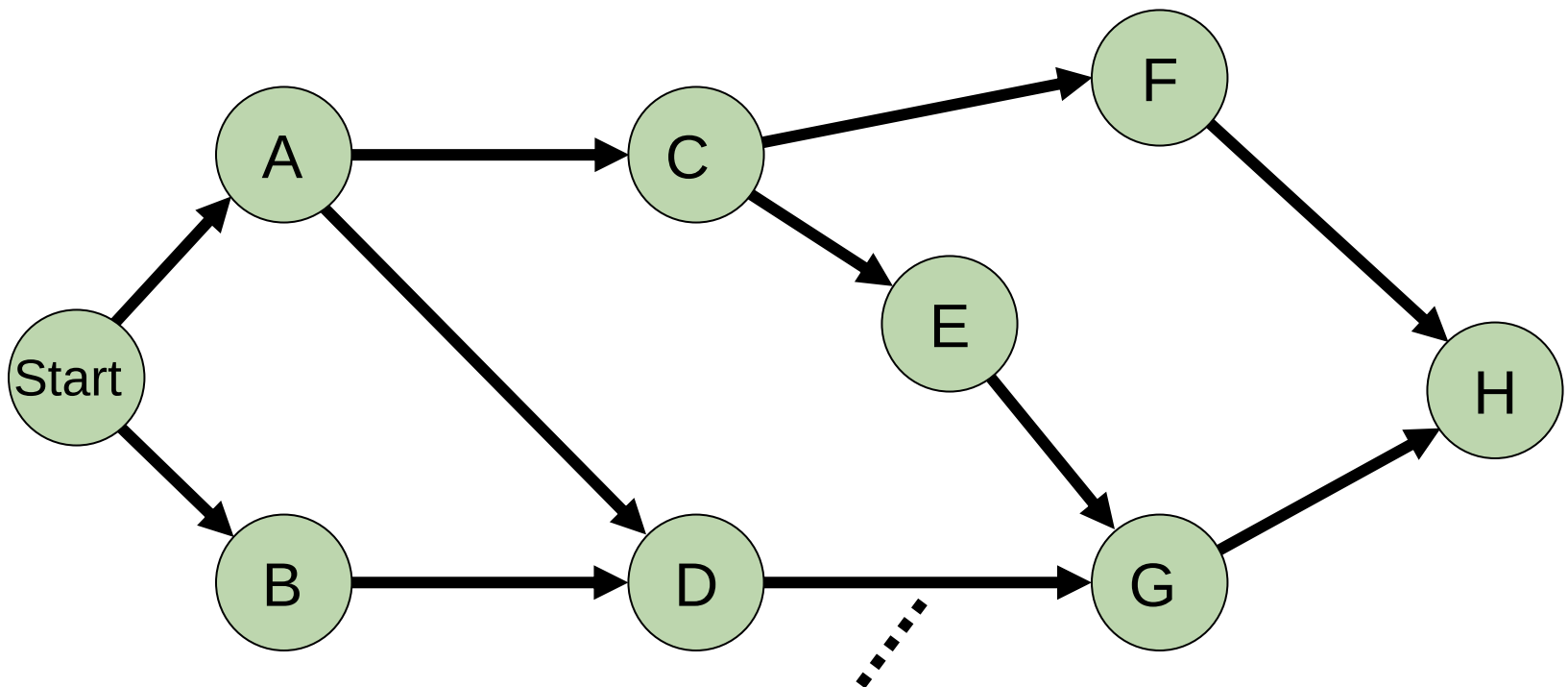


Figure 3.6

AON Network for Milwaukee Paper



Arrows Show Precedence Relationships

Figure 3.7

AOA Network for Milwaukee Paper

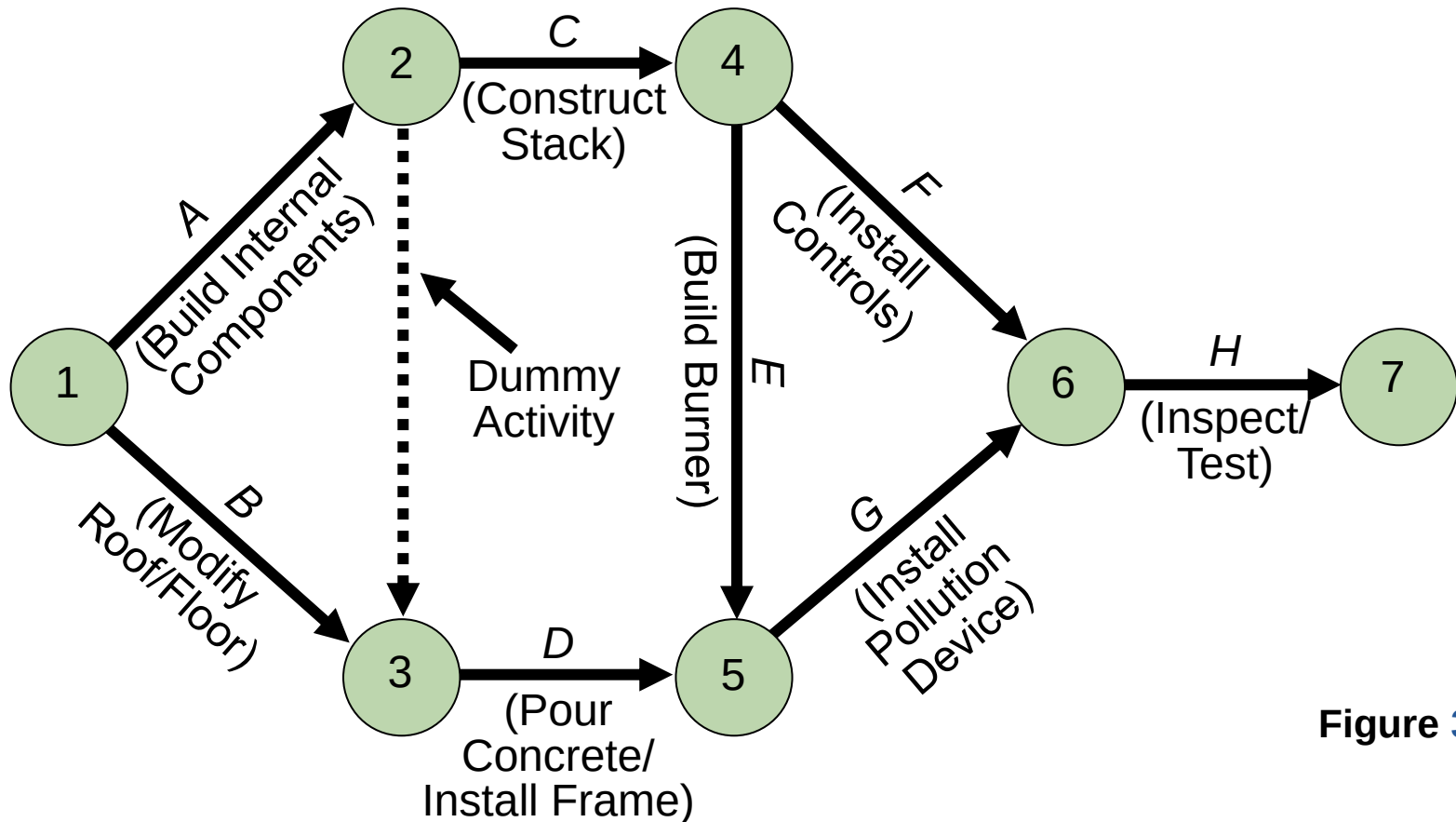


Figure 3.8

Determining the Project Schedule

Perform a Critical Path Analysis

- ▶ The critical path is the longest path through the network
- ▶ The critical path is the shortest time in which the project can be completed
- ▶ Any delay in critical path activities delays the project
- ▶ Critical path activities have no slack time

Determining the Project Schedule

Table 3.2 Time Estimates for Milwaukee Paper Manufacturing

ACTIVITY	DESCRIPTION	TIME (WEEKS)
A	Build internal components	2
B	Modify roof and floor	3
C	Construct collection stack	2
D	Pour concrete and install frame	4
E	Build high-temperature burner	4
F	Install pollution control system	3
G	Install air pollution device	5
H	Inspect and test	2
	Total time (weeks)	25

Determining the Project Schedule

Perform a Critical Path Analysis

Earliest start (ES) = earliest time at which an activity can start, assuming all predecessors have been completed

Earliest finish (EF) = earliest time at which an activity can be finished

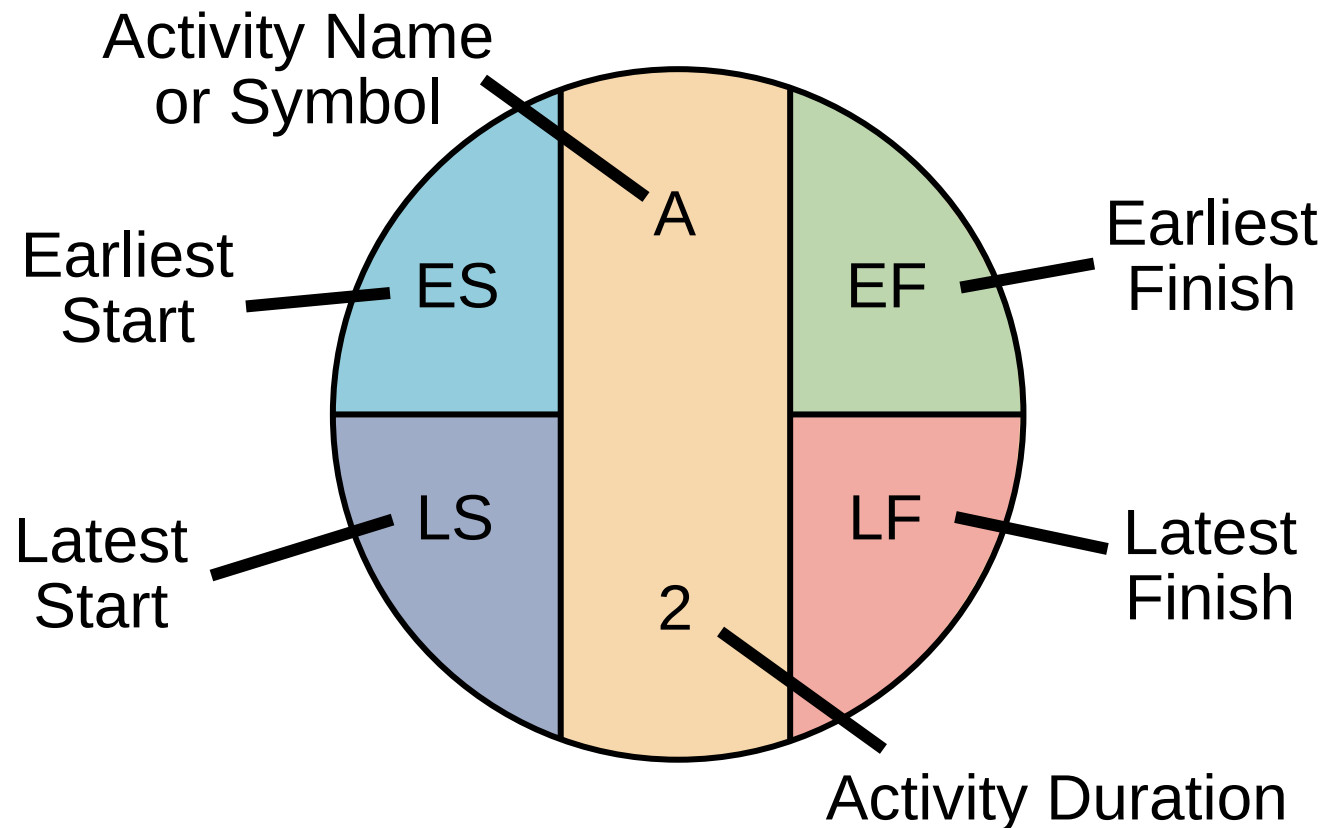
Latest start (LS) = latest time at which an activity can start so as to not delay the completion time of the entire project

Latest finish (LF) = latest time by which an activity has to be finished so as to not delay the completion time of the entire project

Determining the Project Schedule

Activity Format

Figure 3.9



Forward Pass

Begin at starting event and work forward

Earliest Start Time Rule:

- ▶ If an activity has only a single immediate predecessor, its ES equals the EF of the predecessor
- ▶ If an activity has multiple immediate predecessors, its ES is the maximum of all the EF values of its predecessors

$$ES = \text{Max} \{EF \text{ of all immediate predecessors}\}$$

Forward Pass

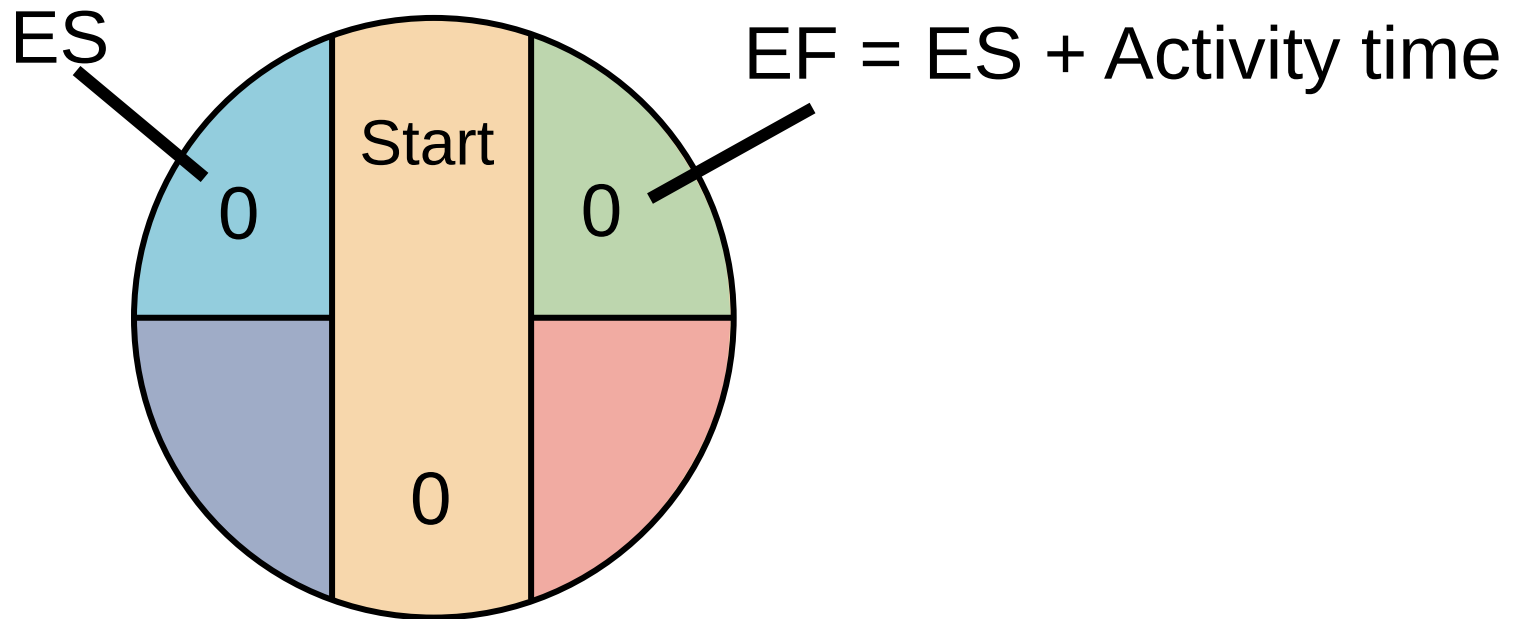
Begin at starting event and work forward

Earliest Finish Time Rule:

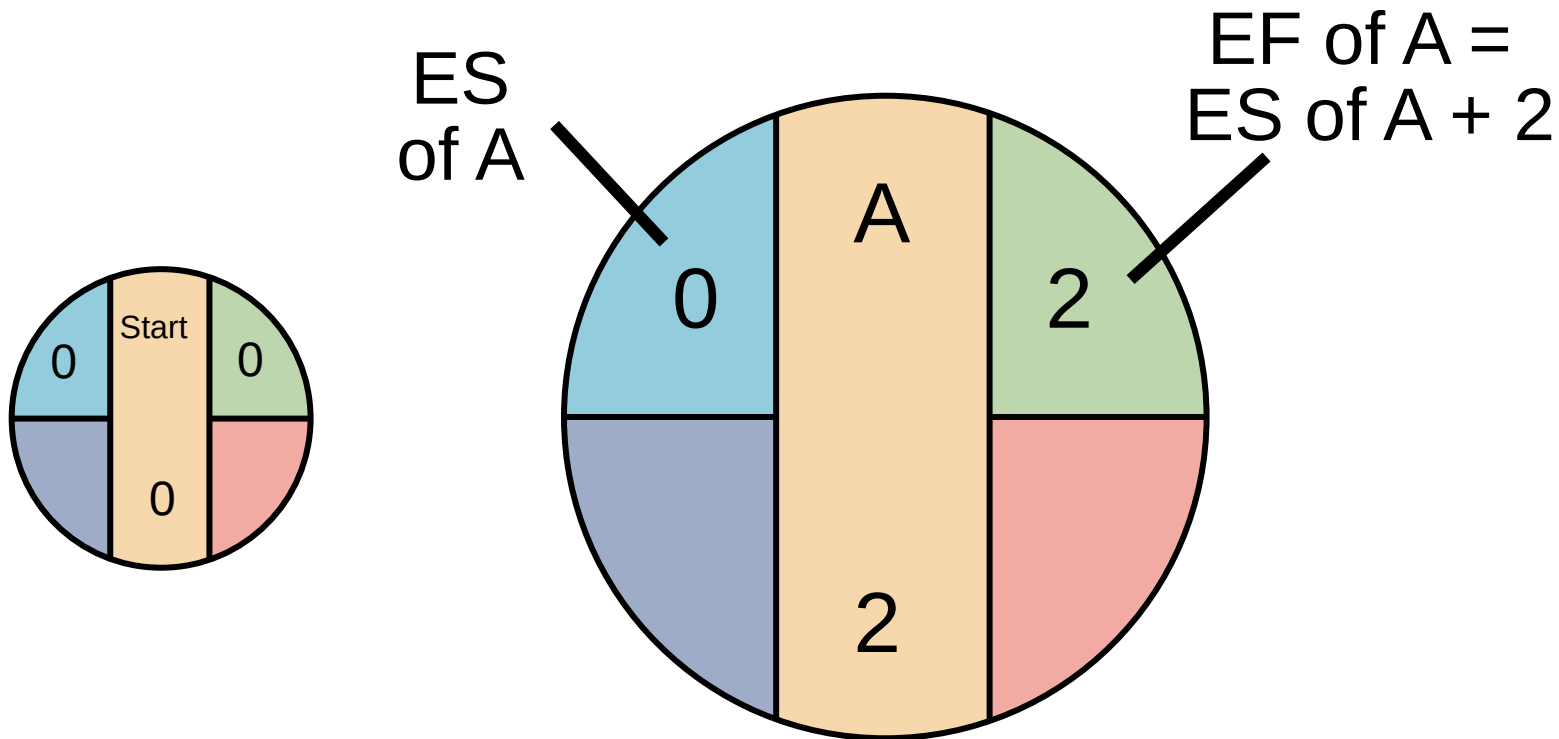
- ▶ The earliest finish time (EF) of an activity is the sum of its earliest start time (ES) and its activity time

$$EF = ES + \text{Activity time}$$

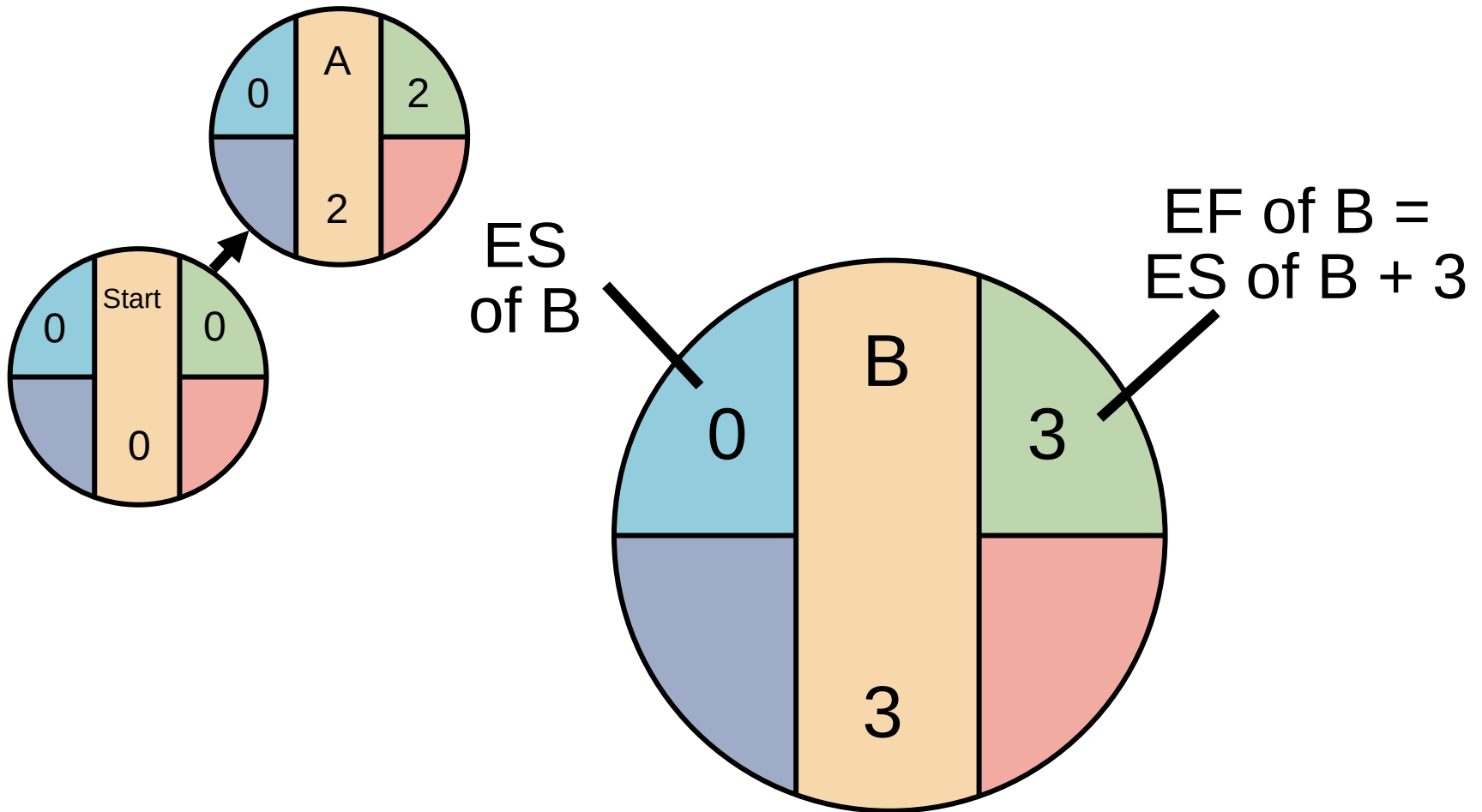
ES/EF Network for Milwaukee Paper



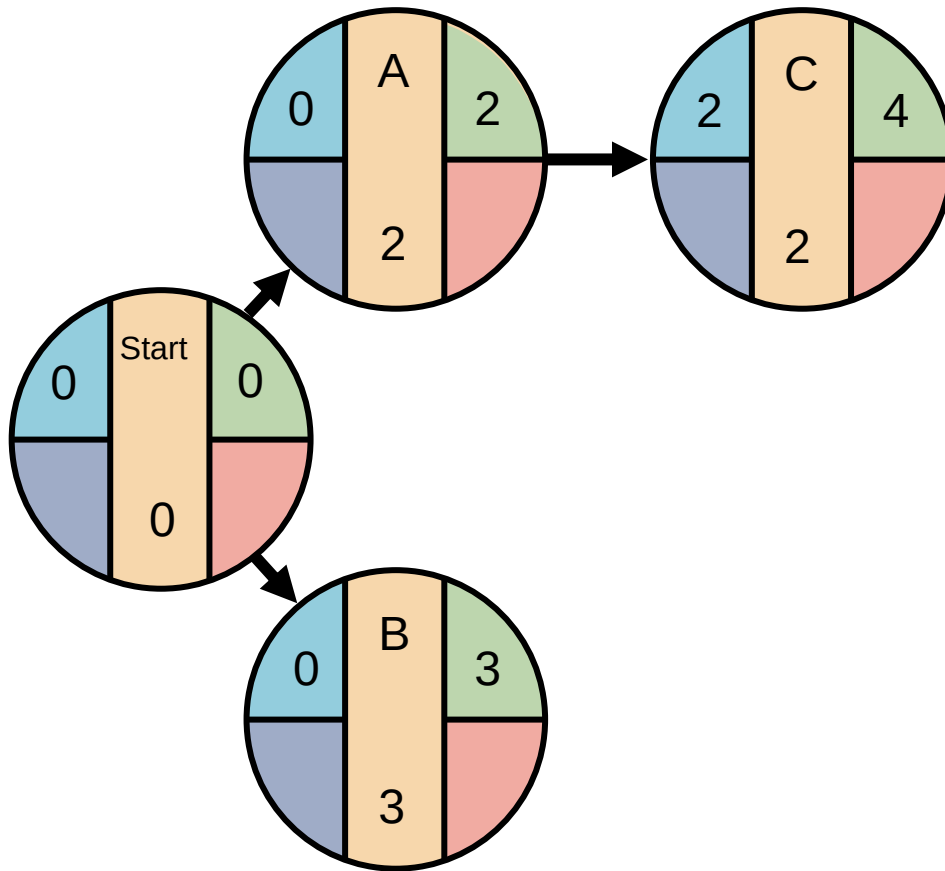
ES/EF Network for Milwaukee Paper



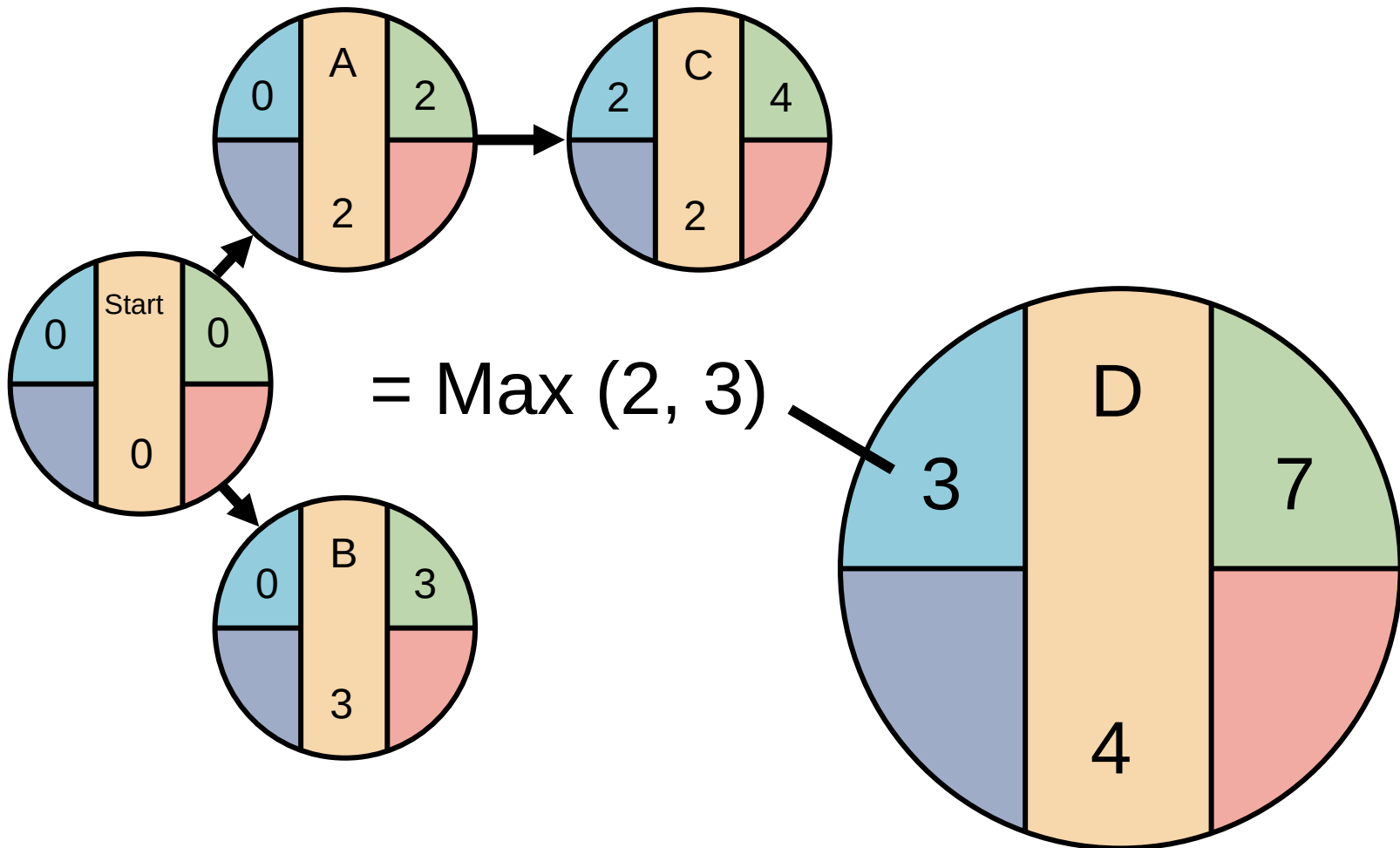
ES/EF Network for Milwaukee Paper



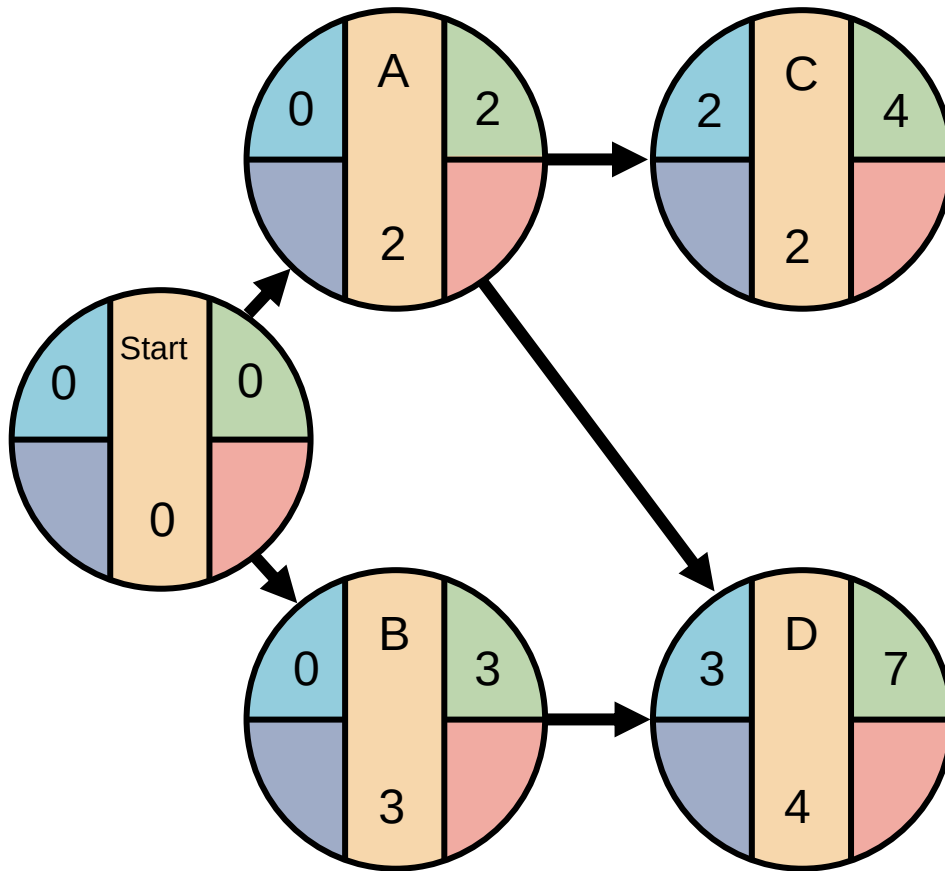
ES/EF Network for Milwaukee Paper



ES/EF Network for Milwaukee Paper



ES/EF Network for Milwaukee Paper



ES/EF Network for Milwaukee Paper

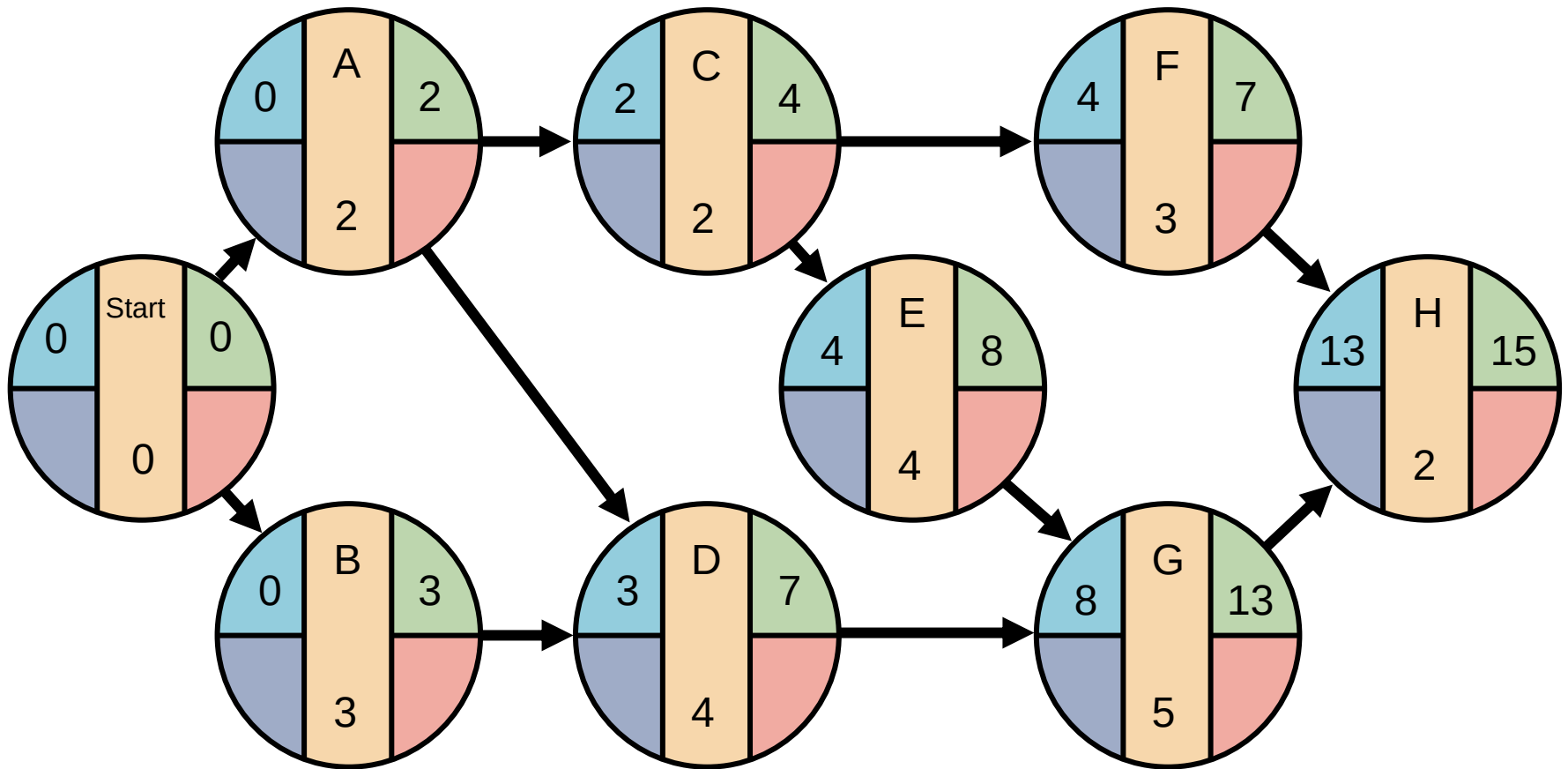


Figure 3.10

Backward Pass

Begin with the last event and work backwards

Latest Finish Time Rule:

- ▶ If an activity is an immediate predecessor for just a single activity, its LF equals the LS of the activity that immediately follows it
- ▶ If an activity is an immediate predecessor to more than one activity, its LF is the minimum of all LS values of all activities that immediately follow it

$LF = \text{Min} \{LS \text{ of all immediate following activities}\}$

Backward Pass

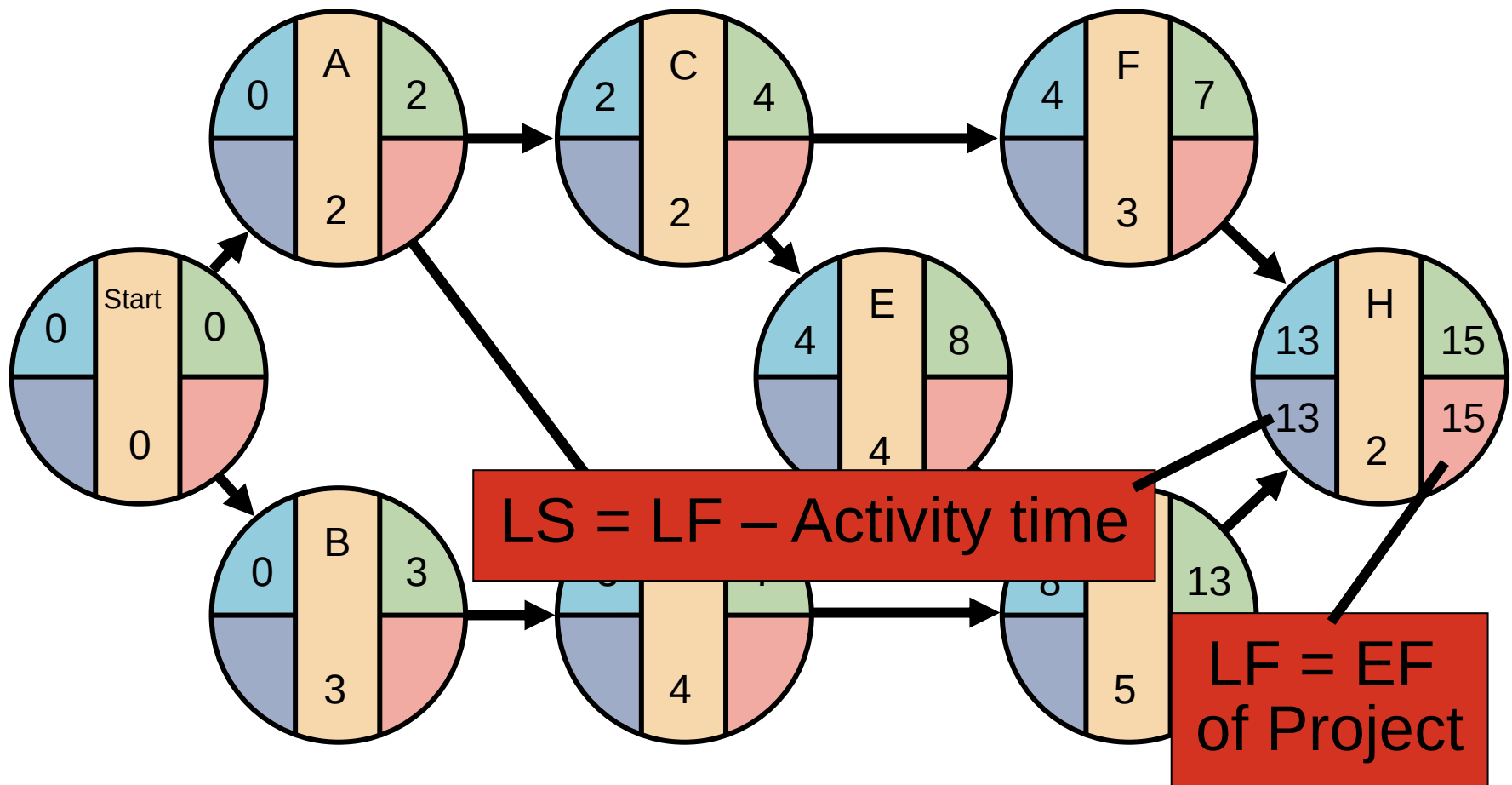
Begin with the last event and work backwards

Latest Start Time Rule:

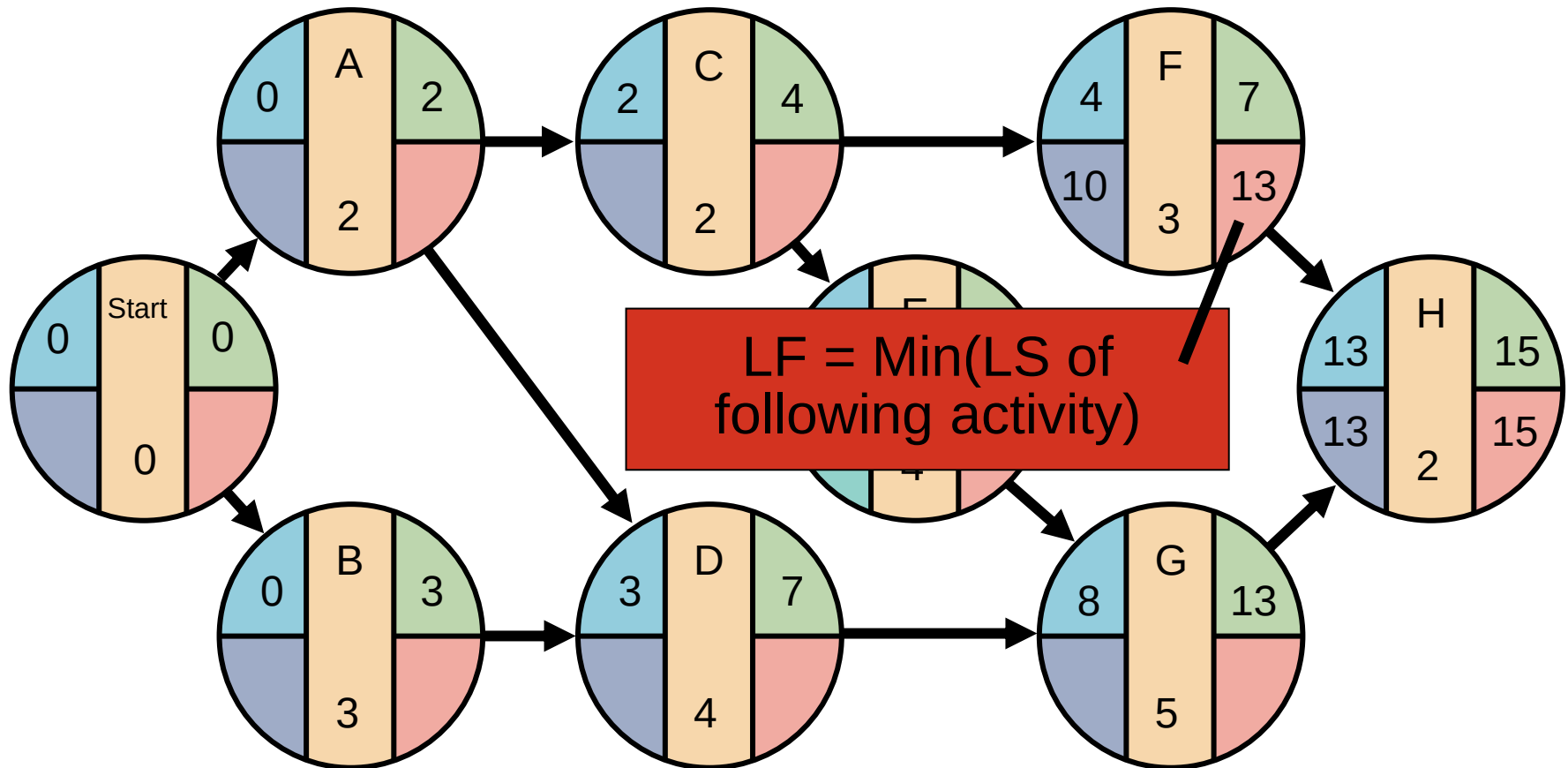
- ▶ The latest start time (LS) of an activity is the difference of its latest finish time (LF) and its activity time

$$LS = LF - \text{Activity time}$$

LS/LF Times for Milwaukee Paper



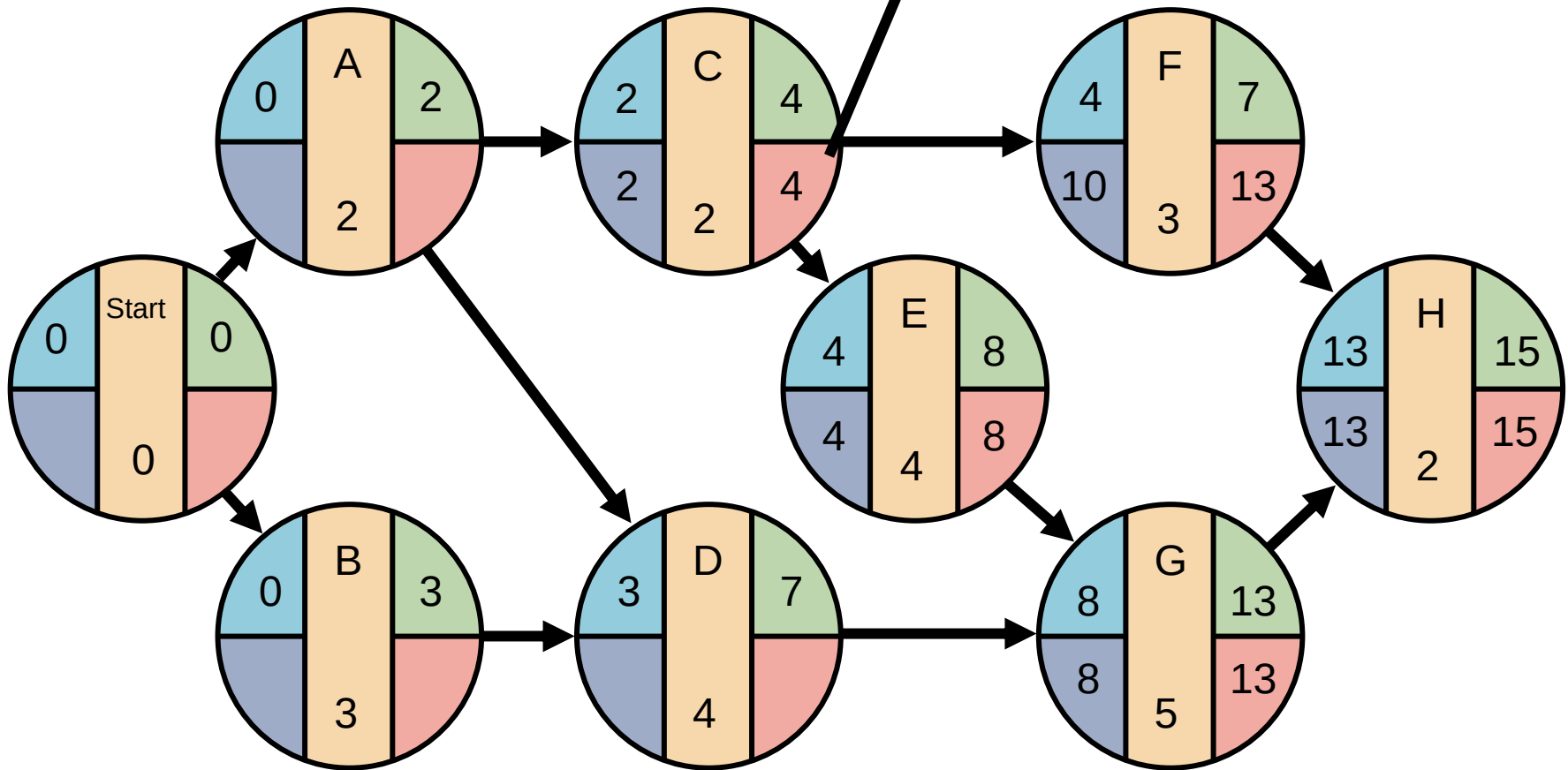
LS/LF Times for Milwaukee Paper



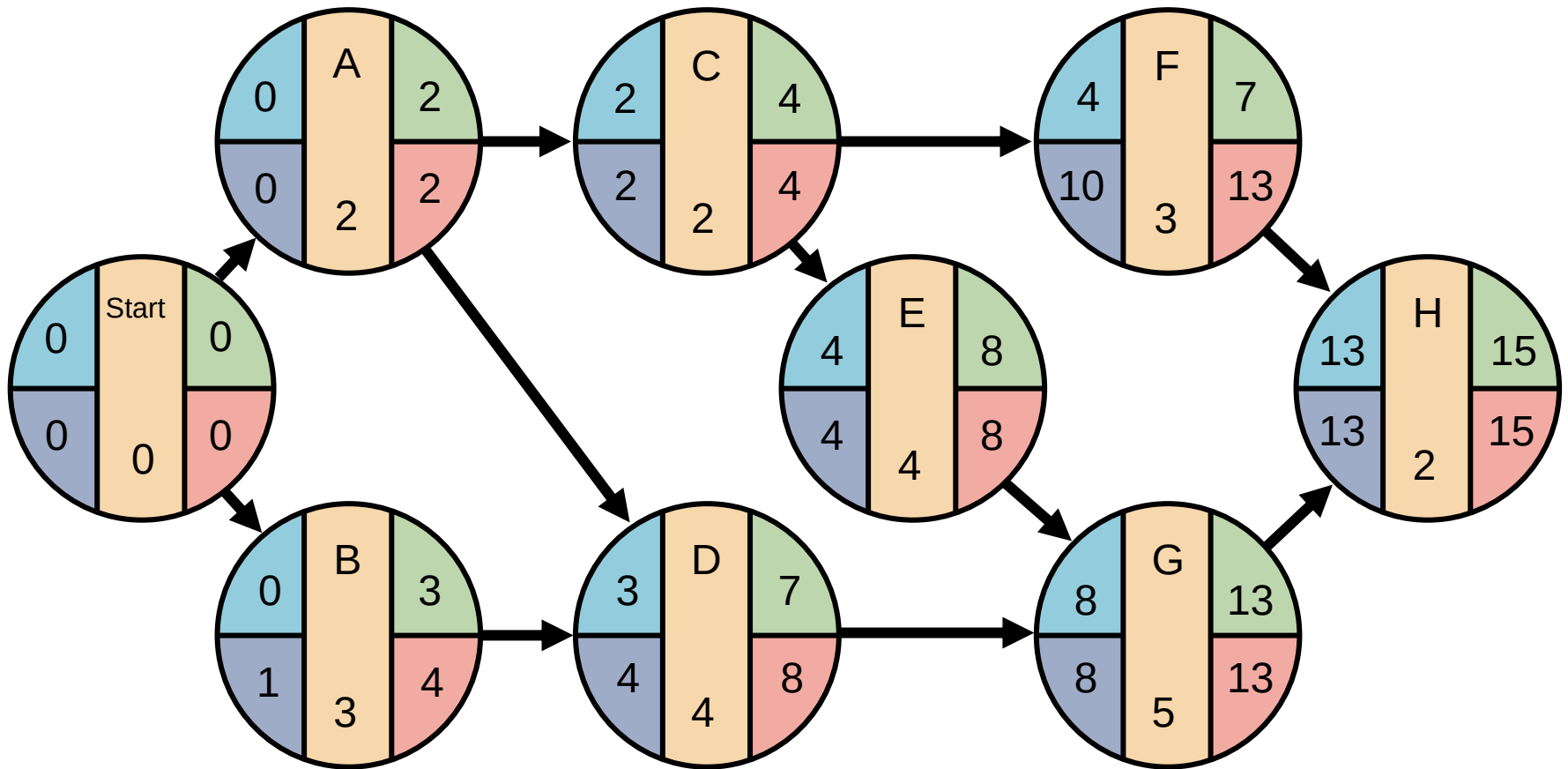
LS/LF Times for

Minimum Cost Super

$$LF = \text{Min}(4, 10)$$



LS/LF Times for Milwaukee Paper



Computing Slack Time

After computing the ES, EF, LS, and LF times for all activities, compute the slack or free time for each activity

- ▶ Slack is the length of time an activity can be delayed without delaying the entire project

$$\text{Slack} = \text{LS} - \text{ES} \quad \text{or} \quad \text{Slack} = \text{LF} - \text{EF}$$

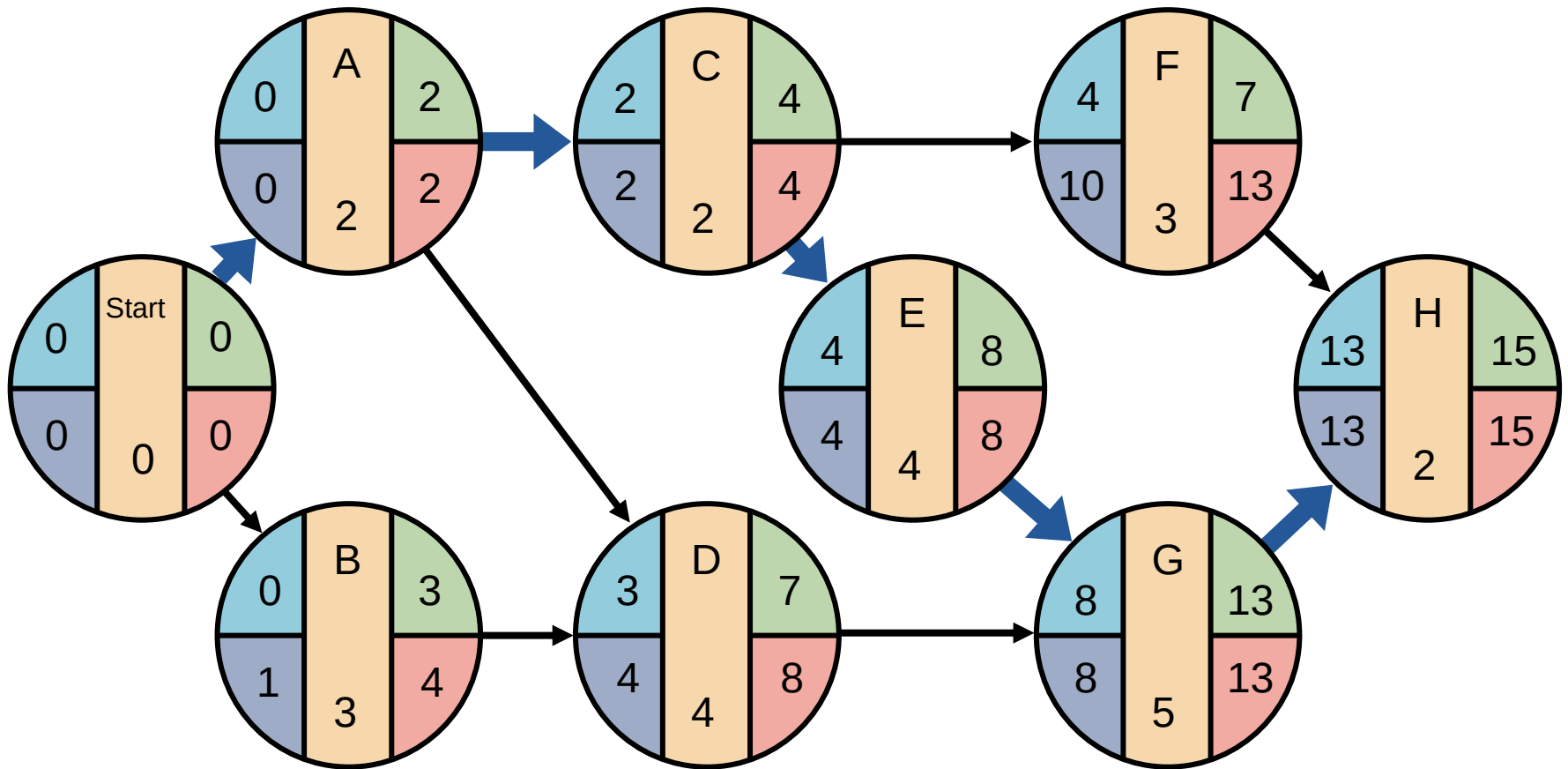
Computing Slack Time

**TABLE
3.3**

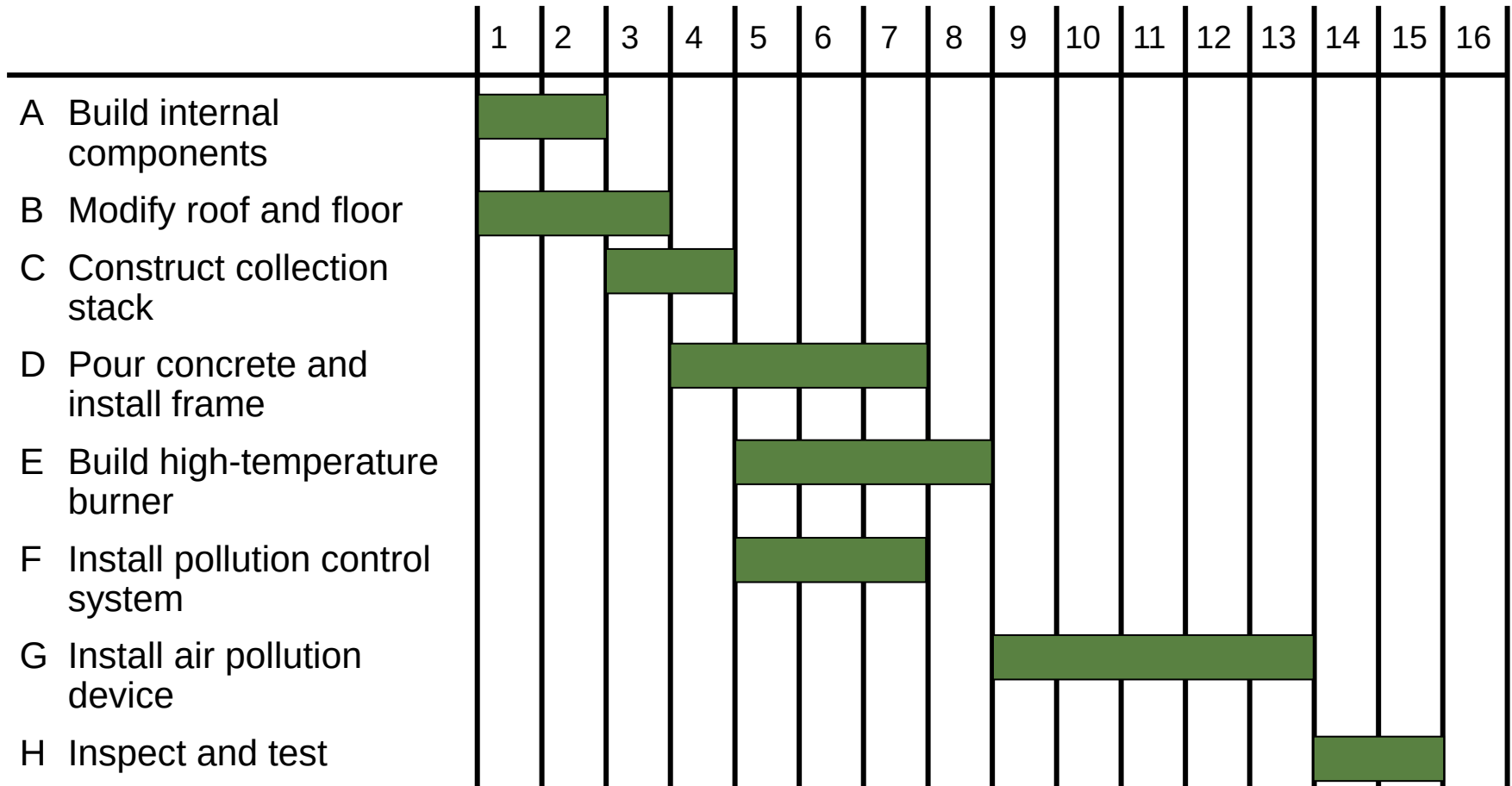
Milwaukee Paper's Schedule and Slack Times

ACTIVITY	EARLIEST START ES	EARLIEST FINISH EF	LATEST START LS	LATEST FINISH LF	SLACK LS – ES	ON CRITICAL PATH
A	0	2	0	2	0	Yes
B	0	3	1	4	1	No
C	2	4	2	4	0	Yes
D	3	7	4	8	1	No
E	4	8	4	8	0	Yes
F	4	7	10	13	6	No
G	8	13	8	13	0	Yes
H	13	15	13	15	0	Yes

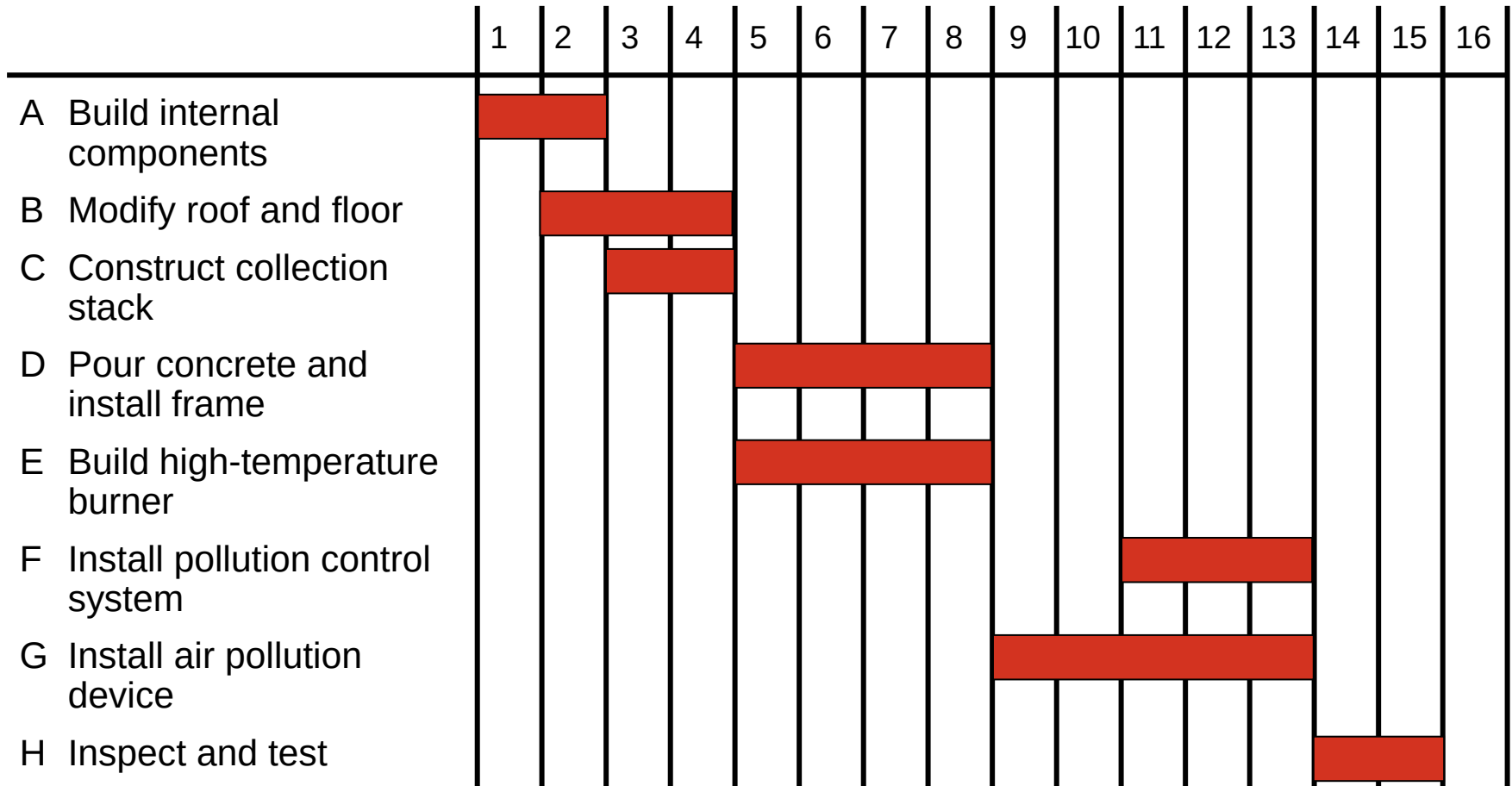
Critical Path for Milwaukee Paper



ES – EF Gantt Chart for Milwaukee Paper



LS – LF Gantt Chart for Milwaukee Paper



Variability in Activity Times

- ▶ CPM assumes we know a fixed time estimate for each activity and there is no variability in activity times
- ▶ PERT uses a probability distribution for activity times to allow for variability

Variability in Activity Times

- ▶ Three time estimates are required
 - ▶ **Optimistic time** (a) – if everything goes according to plan
 - ▶ **Pessimistic time** (b) – assuming very unfavorable conditions
 - ▶ **Most likely time** (m) – most realistic estimate

Variability in Activity Times

Estimate follows beta distribution

Expected time:

$$t = (a + 4m + b)/6$$

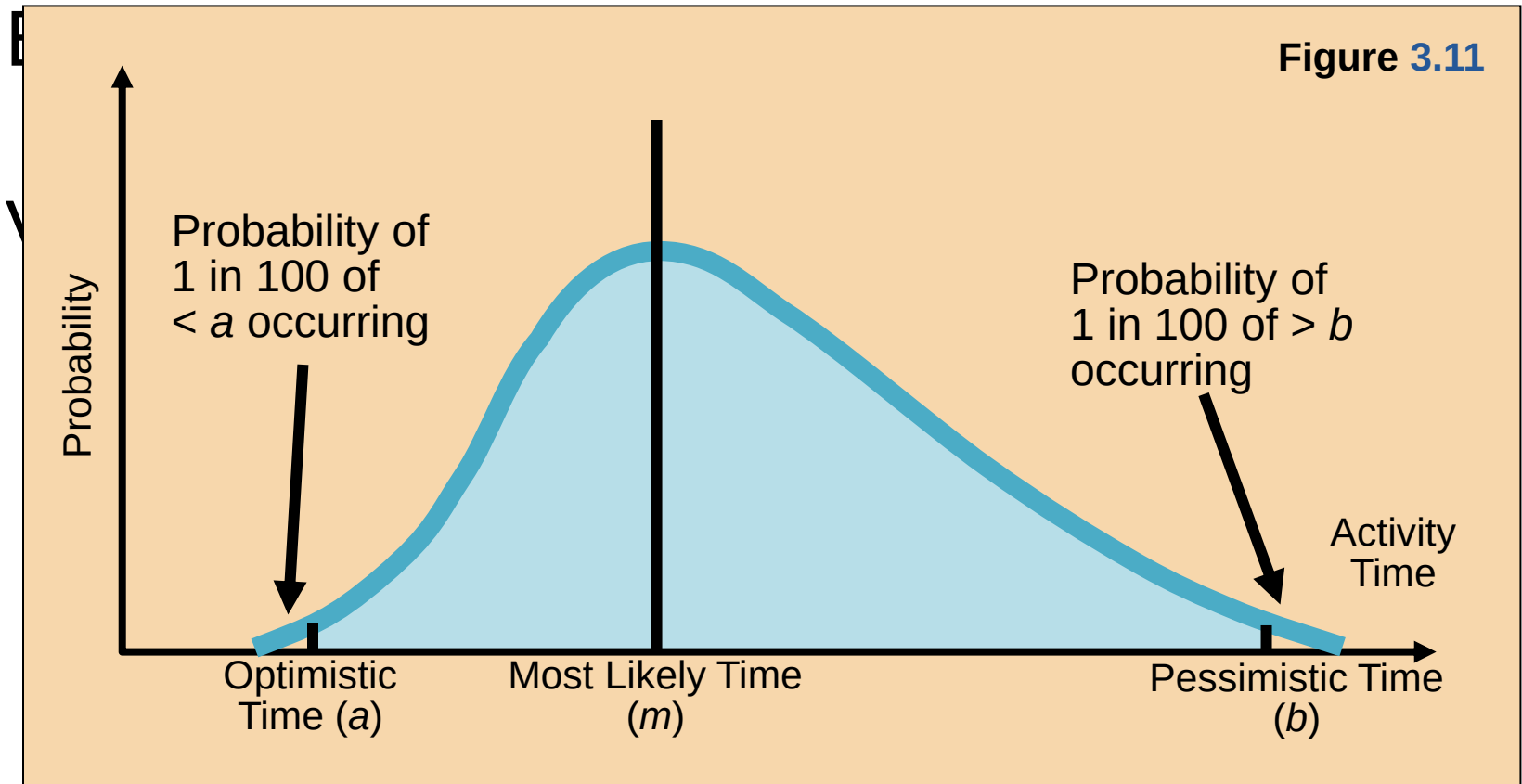
Variance of times:

$$v = [(b - a)/6]^2$$

Variability in Activity Times

Estimate follows beta distribution

Figure 3.11



Computing Variance

TABLE 3.4

Time Estimates (in weeks) for Milwaukee Paper's Project

ACTIVITY	OPTIMISTIC <i>a</i>	MOST LIKELY <i>m</i>	PESSIMISTIC <i>b</i>	EXPECTED TIME $t = (a + 4m + b)/6$	VARIANCE $[(b - a)/6]^2$
A	1	2	3	2	.11
B	2	3	4	3	.11
C	1	2	3	2	.11
D	2	4	6	4	.44
E	1	4	7	4	1.00
F	1	2	9	3	1.78
G	3	4	11	5	1.78
H	1	2	3	2	.11

Probability of Project Completion

Project variance is computed by summing the variances of critical activities

σ_p^2 = Project variance

= $\sum$ (variances of activities
on critical path)

Probability of Project Completion

Project variance is computed by

Project variance

$$\sigma_p^2 = .11 + .11 + 1.00 + 1.78 + .11 = 3.11$$

Project standard deviation

$$\begin{aligned}\sigma_p &= \sqrt{\text{Project variance}} \\ &= \sqrt{3.11} = 1.76 \text{ weeks}\end{aligned}$$

Probability of Project Completion

PERT makes two more assumptions:

- ▶ Total project completion times follow a normal probability distribution
- ▶ Activity times are statistically independent

Probability of Project Completion

Figure 3.12

Standard deviation = 1.76 weeks



Probability of Project Completion

What is the probability this project can be completed on or before the 16 week deadline?

$$\begin{aligned} Z &= \left(\text{Due date} - \text{Expected date of completion} \right) / \sigma_p \\ &= (16 \text{ weeks} - 15 \text{ weeks}) / 1.76 \\ &= 0.57 \end{aligned}$$

Where Z is the number of standard deviations the due date or target date lies from the mean or expected date

Probability of Project Completion

From Appendix I

What
be co
dead

	.00	.01	}}	.07	.08
.1	.50000	.50399		.52790	.53188
.2	.53983	.54380		.56749	.57142
≈					
.5	.69146	.69497		.71566	.71904
.6	.72575	.72907		.74857	.75175

= 0.57

Where Z is the number of standard deviations the due date or target date lies from the mean or expected date

Probability of Project Completion

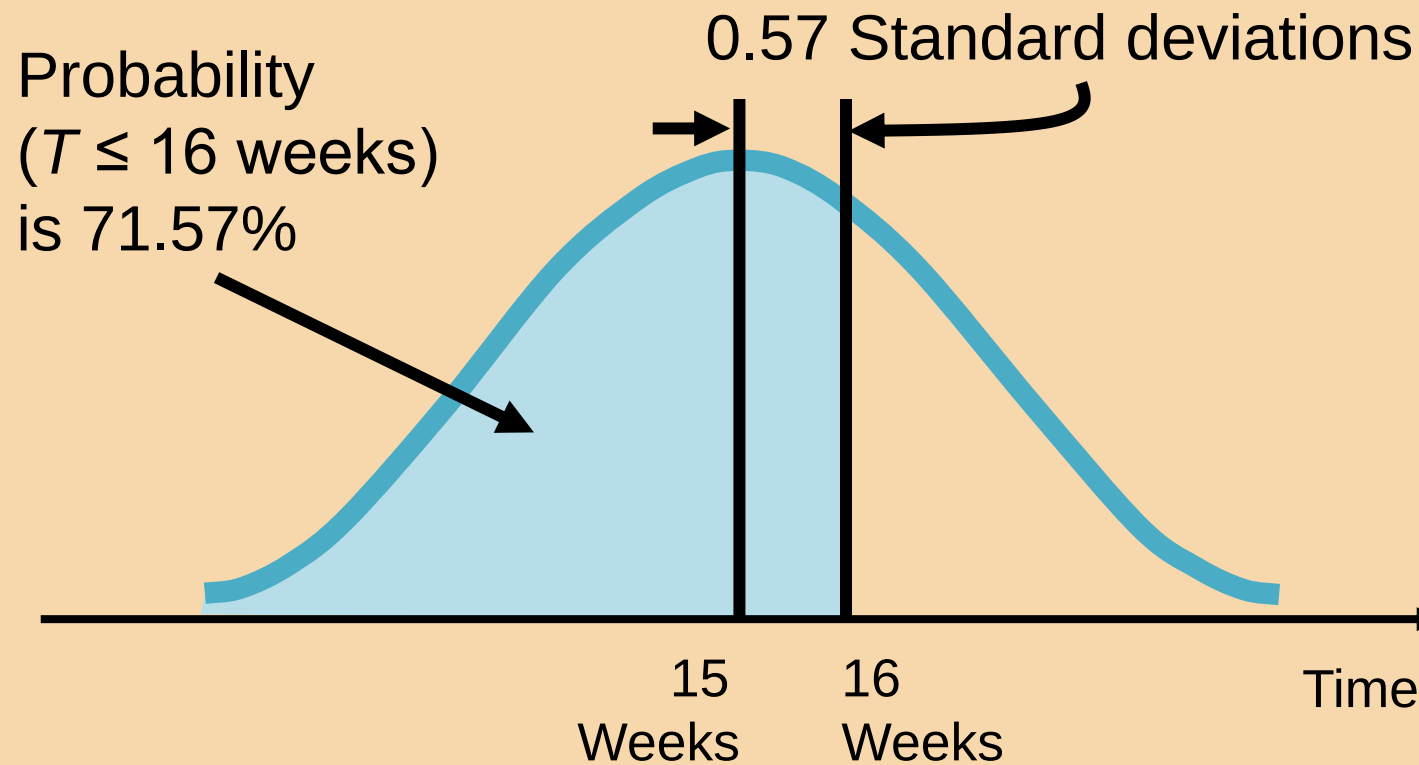


Figure 3.13

Determining Project Completion Time

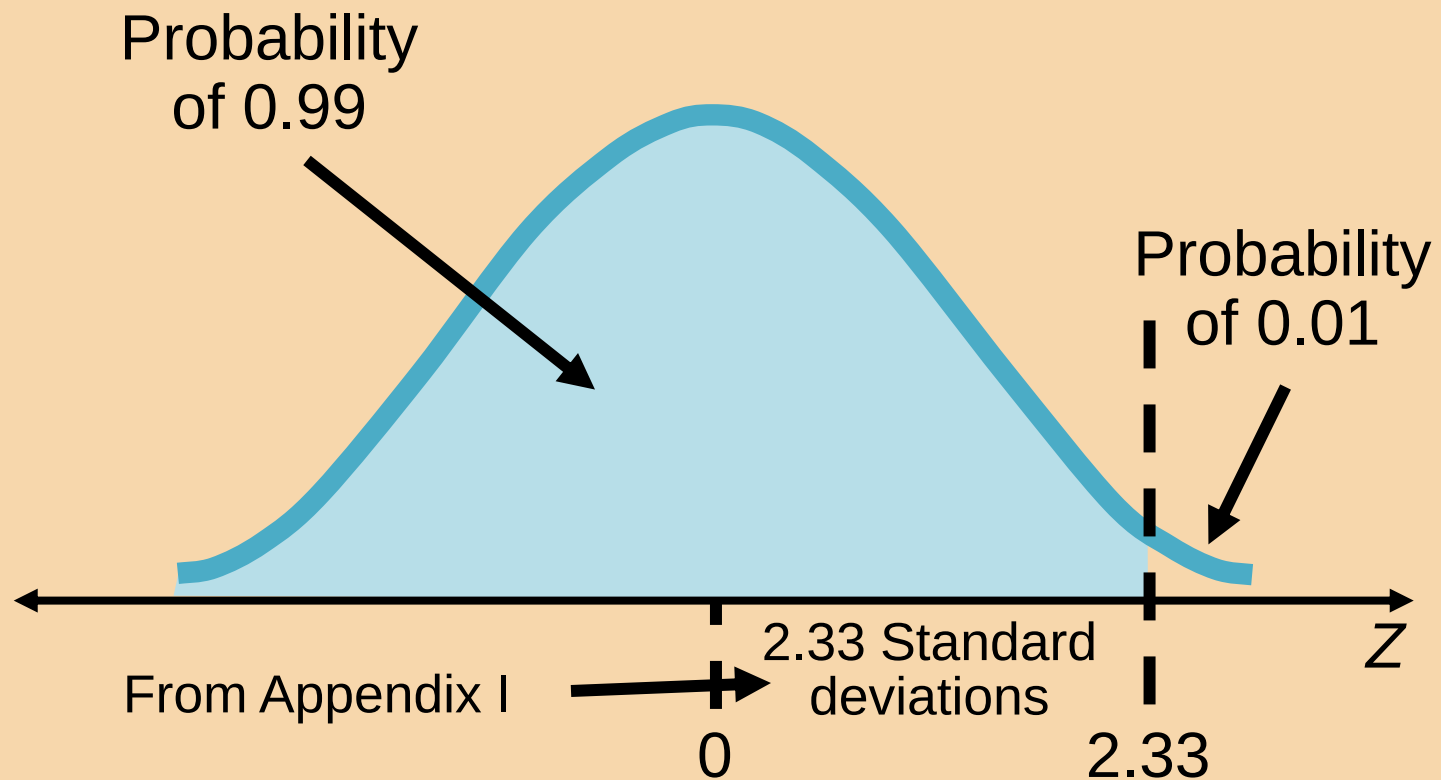


Figure 3.14

Variability of Completion Time for Noncritical Paths

- ▶ Variability of times for activities on noncritical paths must be considered when finding the probability of finishing in a specified time
- ▶ Variation in noncritical activity may cause change in critical path

What Project Management Has Provided So Far

1. The project's expected completion time is 15 weeks
2. There is a 71.57% chance the equipment will be in place by the 16 week deadline
3. Five activities (A, C, E, G, and H) are on the critical path
4. Three activities (B, D, F) are not on the critical path and have slack time
5. A detailed schedule is available

Cost–Time Trade-Offs and Project Crashing

It is not uncommon to face the following situations:

- ▶ The project is behind schedule
- ▶ The completion time has been moved forward

Shortening the duration of the project is called project crashing

Factors to Consider When Crashing a Project

- ▶ The amount by which an activity is crashed is, in fact, permissible
- ▶ Taken together, the shortened activity durations will enable us to finish the project by the due date
- ▶ The total cost of crashing is as small as possible

Steps in Project Crashing

1. Compute the crash cost per time period. If crash costs are linear over time:

$$\text{Crash cost per period} = \frac{(\text{Crash cost} - \text{Normal cost})}{(\text{Normal time} - \text{Crash time})}$$

2. Using current activity times, find the critical path and identify the critical activities

Steps in Project Crashing

3. If there is only one critical path, then select the activity on this critical path that (a) can still be crashed, and (b) has the smallest crash cost per period. If there is more than one critical path, then select one activity from each critical path such that (a) each selected activity can still be crashed, and (b) the total crash cost of all selected activities is the smallest. Note that the same activity may be common to more than one critical path.

Steps in Project Crashing

4. Update all activity times. If the desired due date has been reached, stop. If not, return to Step 2.

Crashing The Project

TABLE 3.5

Normal and Crash Data for Milwaukee Paper Manufacturing

ACTIVITY	TIME (WEEKS)		COST (\$)		CRASH COST PER WEEK (\$)	CRITICAL PATH ?
	NORMAL	CRASH	NORMAL	CRASH		
A	2	1	22,000	22,750	750	Yes
B	3	1	30,000	34,000	2,000	No
C	2	1	26,000	27,000	1,000	Yes
D	4	2	48,000	49,000	1,000	No
E	4	2	56,000	58,000	1,000	Yes
F	3	2	30,000	30,500	500	No
G	5	2	80,000	84,500	1,500	Yes
H	2	1	16,000	19,000	3,000	Yes

Crash and Normal Times and Costs for Activity B

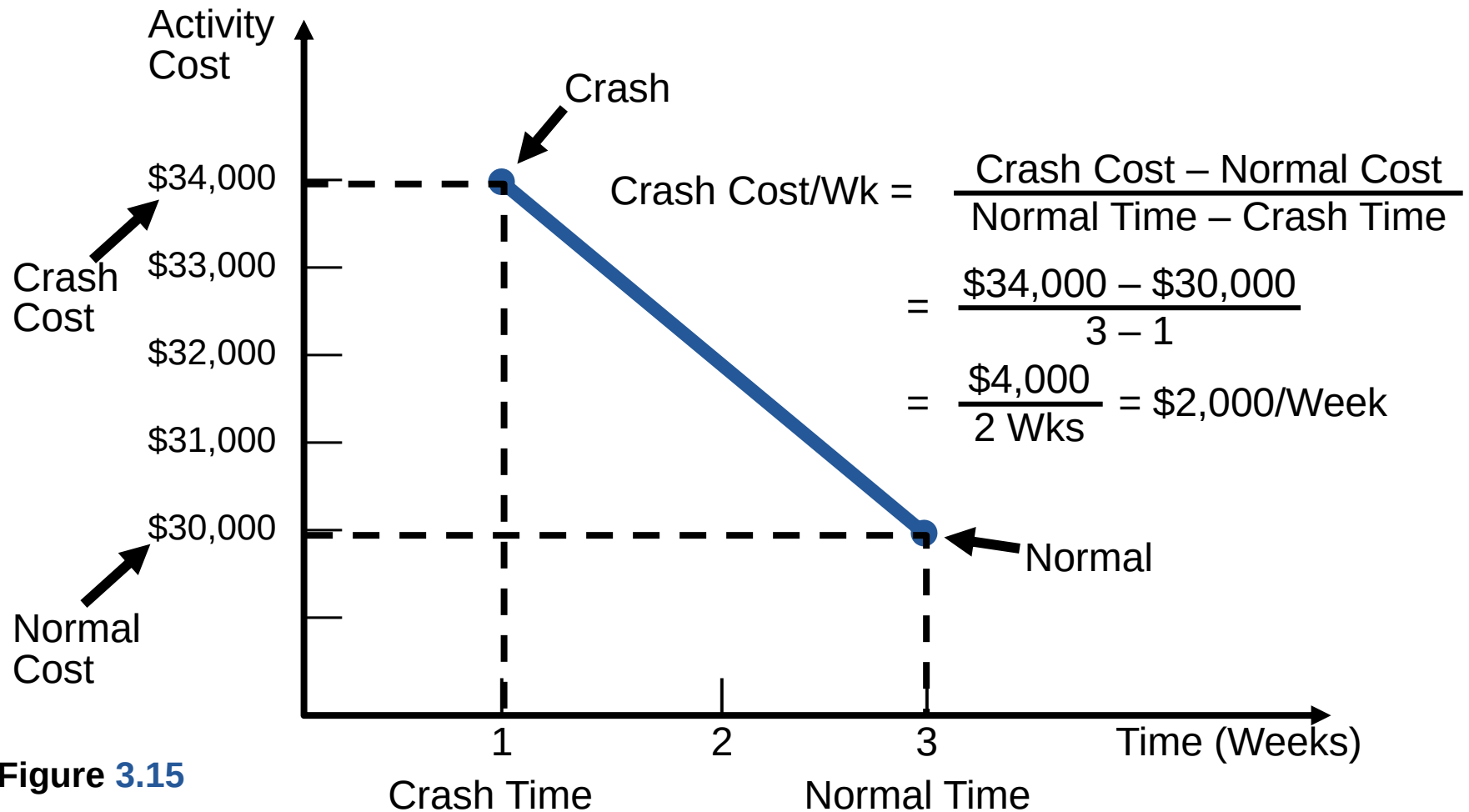
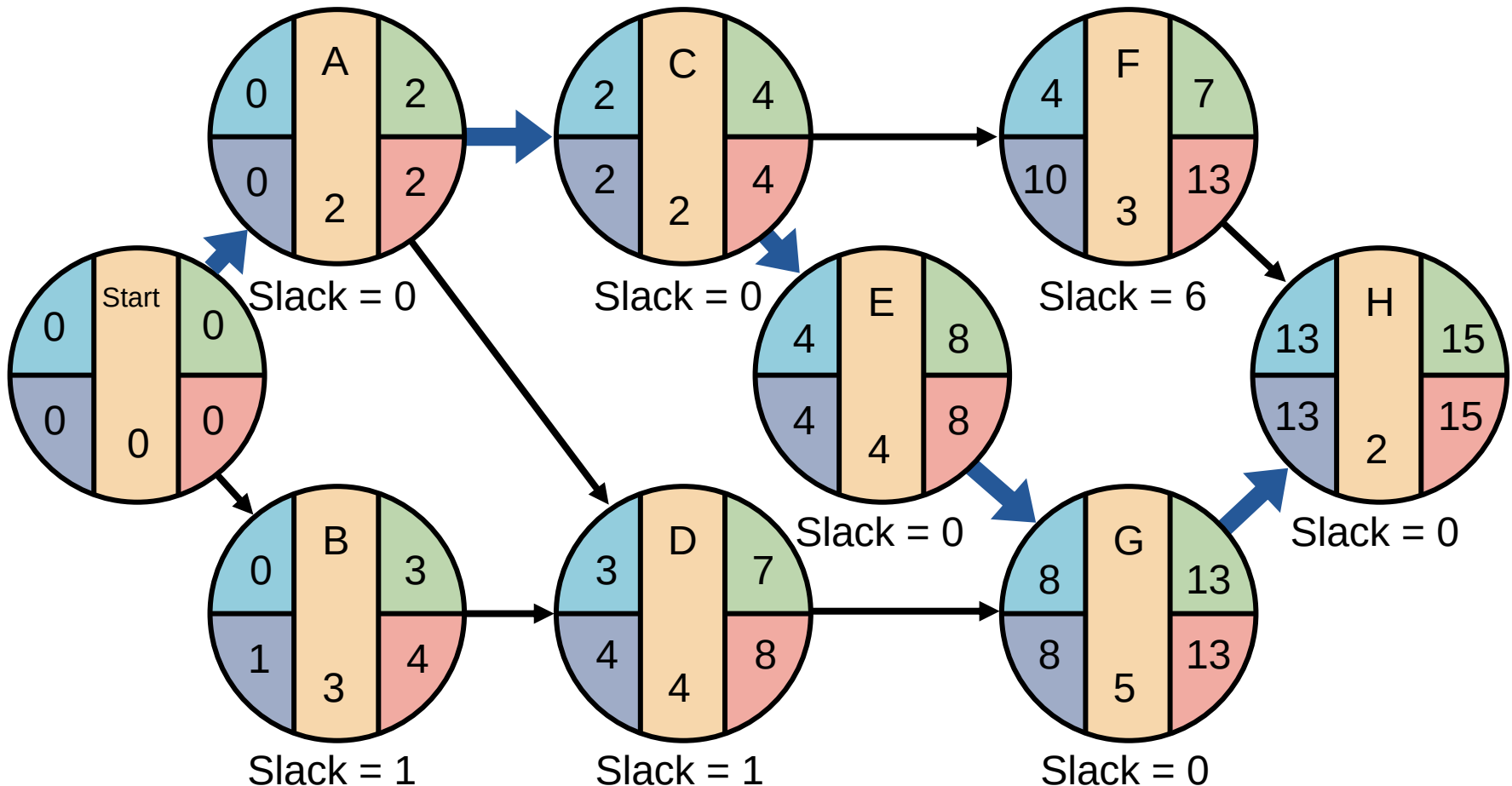


Figure 3.15

Critical Path and Slack Times for Milwaukee Paper

Figure 3.16



Advantages of PERT/CPM

1. Especially useful when scheduling and controlling large projects
2. Straightforward concept and not mathematically complex
3. Graphical networks help highlight relationships among project activities
4. Critical path and slack time analyses help pinpoint activities that need to be closely watched

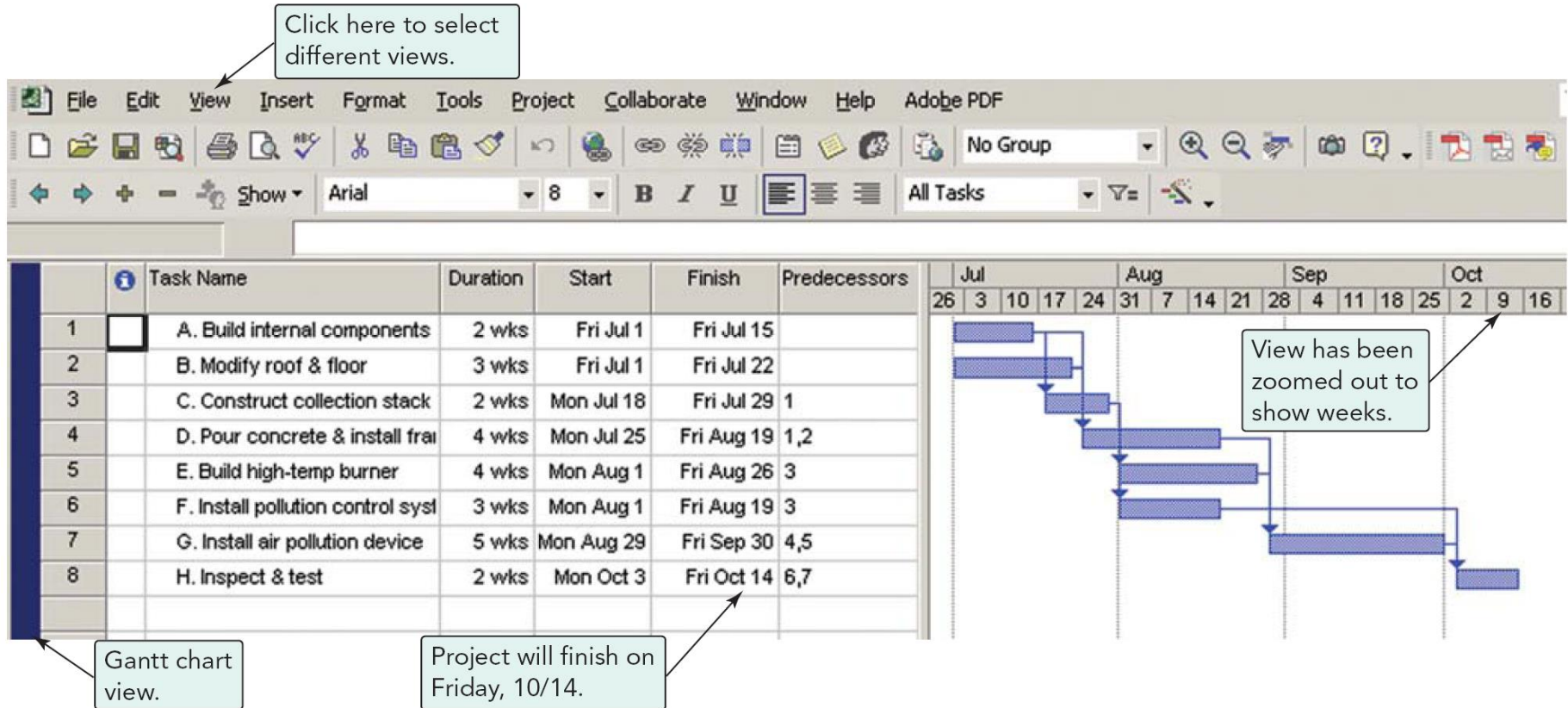
Advantages of PERT/CPM

- 5. Project documentation and graphics point out who is responsible for various activities
- 6. Applicable to a wide variety of projects
- 7. Useful in monitoring not only schedules but costs as well

Limitations of PERT/CPM

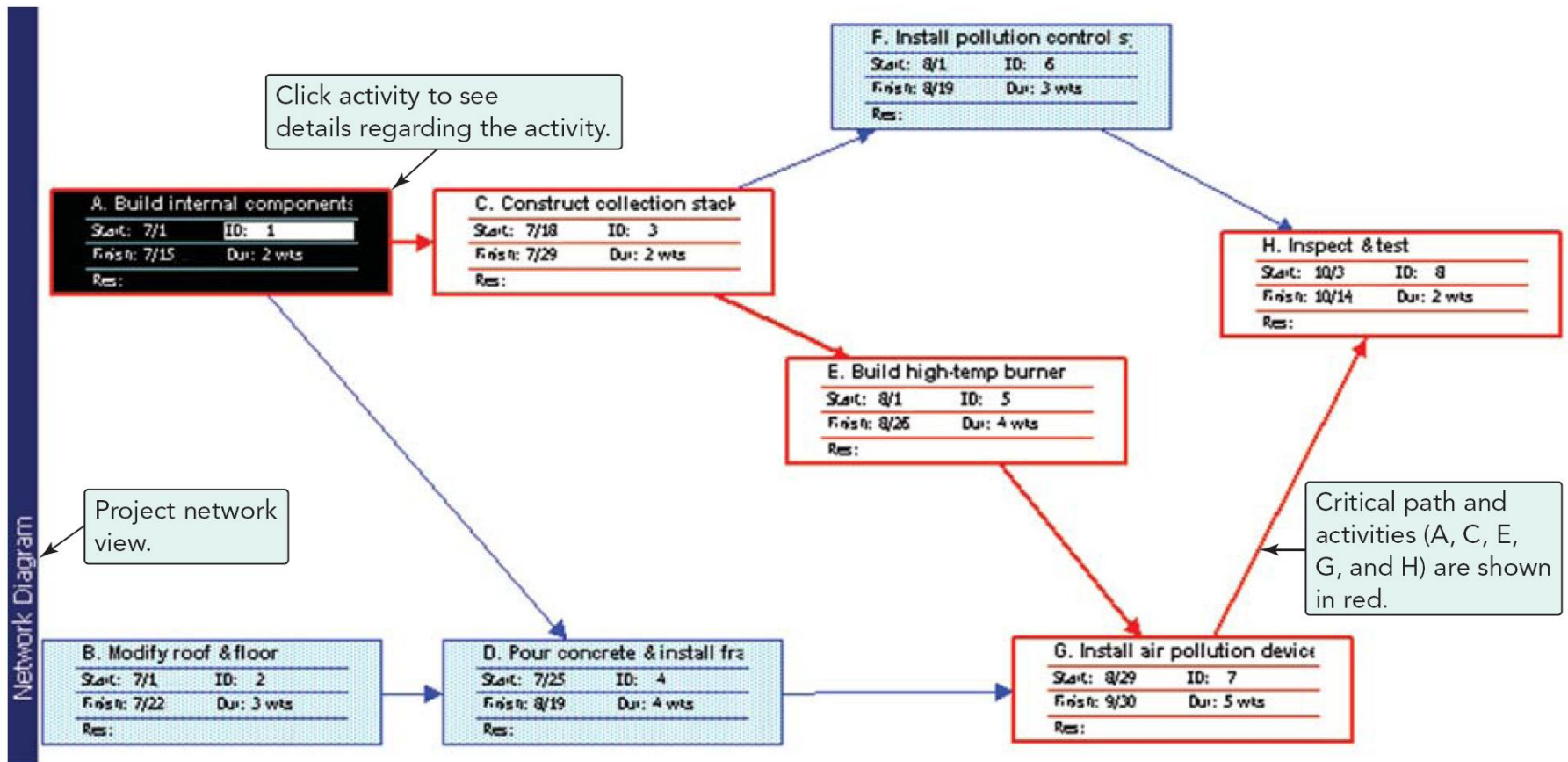
1. Project activities have to be clearly defined, independent, and stable in their relationships
2. Precedence relationships must be specified and networked together
3. Time estimates tend to be subjective and are subject to fudging by managers
4. There is an inherent danger of too much emphasis being placed on the longest, or critical, path

Using Microsoft Project



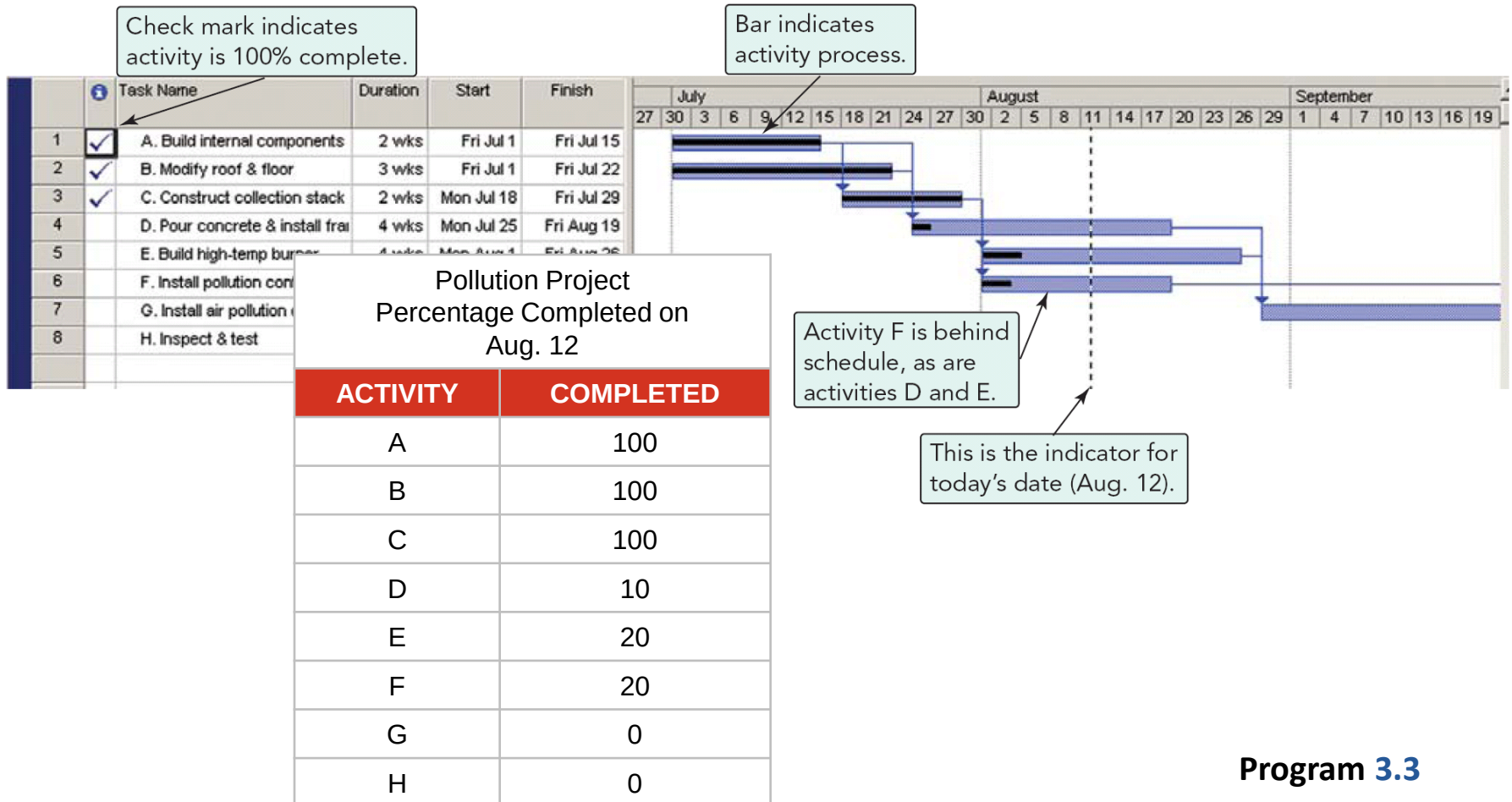
Program 3.1

Using Microsoft Project



Program 3.2

Using Microsoft Project



Program 3.3